



Ministero dell'Università e della Ricerca

Segretariato Generale

Direzione generale della ricerca per la programmazione dei finanziamenti e per l'innovazione tecnologica

Direzione generale per la valutazione e la sicurezza della ricerca

PRIN: PROGETTI DI RICERCA DI RILEVANTE INTERESSE NAZIONALE – Bando 2026
Prot. 2026PST8N3

SECTION A - ADMINISTRATIVE FORM

1. Research Project Title

OPTERA

OPTical satellite multi-tERAbit communication network

2. Duration of the project (months)

36 months

3. Main ERC domain

PE - Physical Sciences and Engineering

4. Other ERC domains

5. ERC subfields

- | | |
|----|--|
| 1. | PE7_6 Communication systems, wireless technology, high-frequency technology |
| 2. | PE7_8 Networks, e.g. communication networks and nodes, Internet of Things, sensor networks, networks of robots |
| 3. | PE7_7 Signal processing |

6. Keywords

1.	Optical communication systems and networks
2.	Terabit non-terrestrial networks
3.	TN/NTN convergence
4.	Coherent WDM transmission
5.	Space data center
6.	Robust broadband communications

7. Principal Investigator

CIVELLI (Surname)	STELLA (Name)
Ricercatore (Qualification)	
01/07/1990 (Date of birth)	CVLSLL90L41D612M (Personal identification code)
Consiglio Nazionale delle Ricerche (Organization)	
	stella.civelli@cnr.it (E-mail address)
stella.civelli@pec.it (PEC)	

Declarations

☒	I declare that I, the substitute and associated PIs have not participated in PRIN 2026 Hybrid call
☒	I declare that I have not received funding under FIS or ERC calls, nor under the high-level research programmes referred to in Article 1, paragraph 4, letter k) of the PRIN 2026 Call
☒	I declare that the time, in percentage terms, dedicated to research activities is: 55 (minimum 20%)
☒	I declare that the substitute PI has a position valid up to 4 years from the date of publication of the PRIN 2026 call.

Current funding and the related time commitment of the PI and associated PIs external to the project, with specific reference to similar projects

n°	Proposal Title	Funding Source	Amount (Euro)	Start Date	End Date	PI/ associated PI Role	Relation to current Proposal	Time, in percentage terms, dedicated to the project	Overlap in scientific content of the project
1.	MASSCON - Garelo	ESA	100.000	15/01/2023	15/09/2026	CNIT unit leader	unrelated	1,00	The two projects are both related to NTN, but focus on different research topics and scientific objectives. The time declared is intended from the submission date on
2.	HANDING OVER - Garelo	ESA	100.000	15/10/2024	15/09/2026	POLITO unit leader	partially related (routing, RRM, handover)	2,00	The ESA project partially overlaps with some activities of WP4, particularly in routing, handover, and radio resource management. However, the ESA project is close to completion (September 2026) and has provided background expertise. The current proposal builds upon these results and extends them toward new scientific objectives and research challenges focusing on all-optical networks. The time declared is intended from the submission date on
3.	STARGATE - Garelo	ESA	115.000	15/06/2026	15/12/2027	POLITO unit leader	unrelated	10,00	The two projects are both related to NTN, but focus on different research topics and scientific objectives. The time declared is intended from the submission date on
4.	MASSCON - Battaglioni	ESA	100.000	15/01/2023	15/09/2026	WP leader	unrelated	1,00	The two projects are both related to NTN, but focus on different research topics and scientific objectives. The time declared is intended from the submission date on
5.	ALLEGRO - Civelli	EU	13.740.977	15/01/2023	15/12/2026	CNR leader (third party CNIT)	unrelated	1,00	The two projects focus on optical network, ALLEGRO focuses on terrestrial network, while OPTERA focuses on satellite networks and NTN/TN convergence. The time declared is intended from the submission date on
TOTAL			14.155.977						

Applications submitted and related time commitment of the PI and associated PIs external to the project

n°	Proposal Title	Funding Source	Funding requested (Euro)	PI/ associated PI Role	Relation to current Proposal	Time, in percentage terms, dedicated to the project
1.	HYPER-3D - Garelo	MIMIT	455.000	participant	partially related: integration between terrestrial and non-terrestrial networks	5,00
2.	Q-VISION - "Quantum Verified Integrated Secure Optical	MUR	100.000	local PI SSSA	unrelated	10,00

	Networks" - Secondini					
3.	QUALIFY - "QUAntum and cIAssical Lasercom Integrated test Facility" - Secondini	Ministero della difesa	248.000	PI	Partially related: testing of optical satellite communication systems (classical and quantum)	15,00
4.	QUASAR - "Quantum Safe Network" - Fase III - Secondini	Ministero della Difesa	63.000	local PI SSSA	unrelated	5,00
	TOTAL		866.000			

Age limits derogation

I intend to benefit from the amount allocated to under 40 PI. For this reason, I declare that both my substitute and I comply with the requirements set out in Article 3, paragraph 9 for PRIN 2026 call for proposals.

The principal investigator and the substitute are both under 40 at the date of the call. They intend to benefit from the amount allocated to under 40 PI.

8. List of research units

n°	Associated Investigator	Position	Organisation	Registered office (address)	Operating office (address) *	e-mail address
1.	CIVELLI Stella	Ricercatore	Consiglio Nazionale delle Ricerche	Piazzale Aldo Moro, 7 - Roma (RM)	City: Pisa Address Via caruso 16, 56122	stella.civelli@cnr.it (adesione completata il 27/05/2026)
2.	SECONDINI Marco	Professore Associato (L. 240/10)	Scuola Superiore di Studi Universitari e Perfezionamento Sant'Anna	P.zza Martiri della Libertà, 33 - PISA (PI)	City: Pisa Address Via Giuseppe Moruzzi, 1, 56124	m.secondini@santannapisa.it (adesione completata il 27/05/2026)
3.	BATTAGLIONI Massimo	Ricercatore a t.d.-t.pieno (L. 79/2022)	Università Politecnica delle MARCHE	P.zza Roma, 22 - ANCONA (AN)	City: Ancona Address via Brecce Bianche 12, 60131	m.battaglioni@staff.univpm.it (adesione completata il 26/05/2026)
4.	GARELLO Roberto	Professore Associato confermato	Politecnico di TORINO	Corso Duca degli Abruzzi, 24 - TORINO (TO)	City: torino Address corso duca degli abruzz 24, 10129	roberto.garello@polito.it (adesione completata il 25/05/2026)

9. Substitute Principal Investigator (PI)* (To be identified among one of the associated PIs participating in the project or from members of one of the PIs team)

BATTAGLIONI (Surname)	MASSIMO (Name)
Ricercatore a t.d.-t.pieno (L. 79/2022) (Qualification)	
03/08/1991 (Date of birth)	BTTMSM91M03E783W (Personal identification code)
Università Politecnica delle MARCHE (Organization)	
	m.battaglioni@staff.univpm.it (E-mail address)
massimo.battaglioni@pec.it (PEC)	

10. Proposal abstract

OPTERA satellite multi-terAbit communication network (OPTERA) aims to design and validate (TRL 4) a high-capacity space-ground optical coherent communication network. The project addresses the capacity limits of radio-frequency (RF) satellite systems with an optical infrastructure, exploiting the advantages of optical links (high capacity, narrow beams, reduced interference and spectrum scalability, mature fiber technology), while addressing the main free-space optics (FSO) impairments (e.g., attenuation, turbulence, pointing and tracking (PAT) errors, cloud blockage).

OPTERA focuses on two use cases: resilient broadband and service continuity for disaster recovery, remote areas and terrestrial/non-terrestrial network (TN/NTN) convergence; space data centers for artificial intelligence (AI) computing and always-connected Earth observation (EO) satellites. OPTERA advances the state of the art by integrating coherent optical wavelength division multiplexing (WDM) transmission, advanced and adaptive digital signal processing (DSP), transparent forwarding, and programmable networking for space-ground systems. OPTERA is validated in a laboratory network with multi-hop links, commercial and proprietary components, and FSO impairment emulation. OPTERA targets 10 Tb/s, with laboratory validation up to 2.5 Tb/s, and service availability above 99%.

OPTERA is organized into five work packages (WPs) and milestones (MSs), and thirteen task objectives and key performance indicators (KPIs). WP1 manages coordination, dissemination, and exploitation. WP2 defines requirements and architecture. WP3 designs the WDM coherent optical system: channel models for FSO links, end-to-end optical system including multi-hop transparent links, and advanced DSP. WP4 designs the programmable networking including time-aware routing, handover, link switching, resource allocation, and telemetry-driven source routing framework. WP5 implements the testbed and experimentally demonstrates the complete communication network. MSs and KPIs enable a staged validation strategy that connects physical-layer innovation with network-layer operation and laboratory validation. Risk mitigation strategies are included to support the project success. OPTERA brings together four units across three cities, combining complementary and multidisciplinary expertise in physical-layer design, advanced DSP and coding, optical fiber communication, satellite networking, network control, and optical laboratory demonstrations.

OPTERA is aligned with major national, European, and global strategic priorities for space economy, continuous and resilient connectivity for isolated areas and emergency situations, national sovereignty, and data transmission and processing. Dissemination and exploitation activities will boost the impact on society through advancement of knowledge, stakeholder engagement, industrial uptake, and development of simulation tools and experimental platforms for future NTN research.

11. Total cost of the research project identified by items

Research Unit	item A.1	item A.2.1	item B	item C	item D	item E	Total
CIVELLI Stella	233.805	126.000	161.912	0	0	2.800	524.517
SECONDINI Marco	143.375	124.667	120.619	101.000	50.000	2.800	542.461
BATTAGLIONI Massimo	76.500	110.001	83.925	0	15.000	21.073	306.499
GARELLO Roberto	139.125	150.000	130.106	0	0	0	419.231

Total	592.805	510.668	496.562	101.000	65.000	26.673	1.792.708
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N.B. The fields indicated in gray will be filled in automatically

- **Item A.1:** enhancement of months/person of permanent and temporary employees
- **Item A.2.1:** cost of contracts of non-employees, specifically to recruit
- **Item B:** overhead (calculated as 45% of the sum of A.1 and A.2.1)
- **Item C:** cost of equipment and tools
- **Item D:** cost of consulting and other services
- **Item E:** other operating costs

SECTION B - PROPOSAL SUMMARY AND TRACK RECORD

1. SYNOPSIS

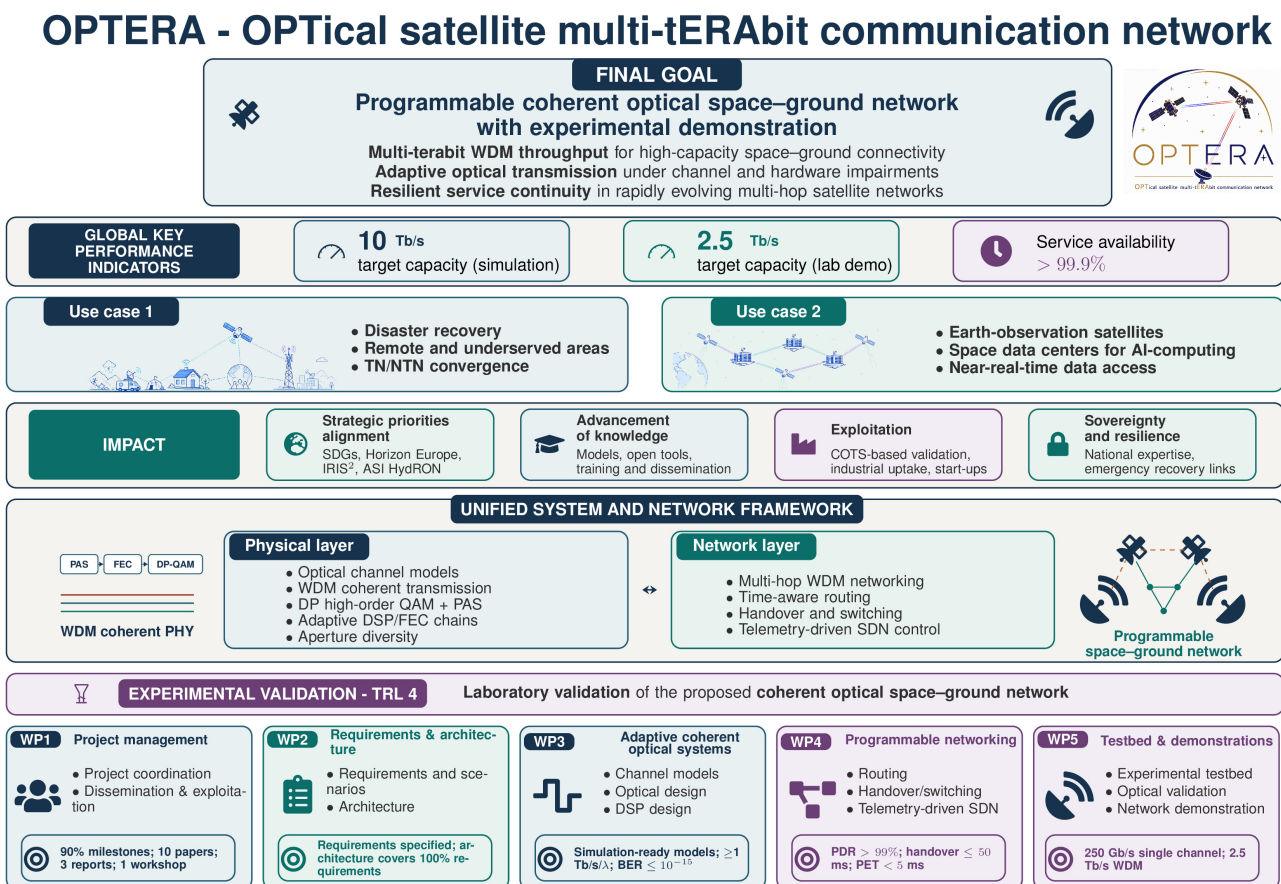


Fig 1: OPTERA in a nutshell

Impact and relevance

Optical satellite multi-tERAbit communication network (OPTERA) aims to design and experimentally validate (TRL 4, validation in a laboratory environment) a programmable network for space-ground communications based on coherent optical systems (Fig 1). The project addresses a strategic challenge: enabling an optical non-terrestrial infrastructure capable of overcoming both the capacity limits of radio-frequency (RF) communications and the bottlenecks of emerging optical links, hampered by free-space optical (FSO) impairments and mostly point-to-point. Its final goal is a scalable end-to-end coherent optical space-ground network with multi-terabit wavelength division multiplexing (WDM) aggregate throughput, dynamic adaptation to FSO impairments, and resilient service continuity under rapidly evolving multi-hop satellite network conditions.

OPTERA is relevant because low-Earth orbit (LEO) constellations, terrestrial/non-terrestrial networks (TN/NTN) convergence and resilient connectivity are central to digital infrastructures. The project targets two use cases (UCs): UC1: resilient broadband and service continuity for disaster recovery, remote and underserved areas, and TN/NTN convergence; and UC2: space data centers for artificial intelligence (AI) computing and always-connected Earth observation (EO) satellites, where optical links enable rapid data transfer, distributed processing, and continuous access to space-generated information.

OPTERA brings the advantages of optical technologies to space by integrating high-capacity coherent optical communications with

advanced digital signal processing (DSP), LEO routing, handover and switching algorithms, and telemetry-driven software-defined networking (SDN). OPTERA demonstrates an integrated laboratory network with transparent multi-hop links, using WDM coherent systems and commercial off-the-shelf (COTS) components. These activities bridge optical communications, satellite networking, DSP, and coding.

OPTERA supports emergency communications, disaster recovery, connectivity in remote areas, EO, AI-driven data processing and public services. For hazard-prone Italy, OPTERA contributes to resilient national connectivity and strengthens strategic competences in space communications and optical networking. OPTERA is aligned with European and national priorities in secure connectivity, Space Economy, technological sovereignty, TN/NTN convergence, IRIS2 flagship, HydRON project (ASI, ESA) and Horizon work programs on next-generation satellite communications. OPTERA addresses sustainable development goal (SDG) "Industry, Innovation and Infrastructure" and indirectly supports SDGs related to health, education, economic growth, digital inclusion, and strong institutions. OPTERA also contributes to the advancement of knowledge through publications, public technical reports, open software tools, and a dedicated scientific workshop. It supports the training of PhD students and early-career researchers in a strategic technological domain. The project-generated intellectual property may support future start-up initiatives.

Objectives beyond the state of the art

The global OPTERA objective is to design and experimentally validate (TRL 4) a scalable 10 Tb/s space-ground optical network, with laboratory validation up to 2.5 Tb/s, and laboratory validated end-to-end service availability above 99.9% under dynamic multi-hop networking conditions. State of the art links achieve <1Tb/s (NASA TBIRD experiment: 200Gb/s downlink, Starlink claim: 200Gb/s intersatellite). OPTERA advances satellite communications targeting an optical space network that, compared to RF links, offers higher capacity, highly directional narrow beams, reduced interference, improved spectrum scalability, and absence of spectrum regulations, while leveraging mature terrestrial optical technologies. Additionally, OPTERA provides innovative solutions to address FSO impairments (e.g., atmospheric attenuation, turbulence, pointing-acquisition-tracking (PAT) errors, phase noise, Doppler, cloud blockage), dynamic network topology, and onboard resource constraints. OPTERA advances the state of the art through a unified optical space-ground communication chain: from coherent optical WDM transmission with advanced DSP to routing, handover, telemetry-driven SDN control and validation at TRL 4.

The main innovation of OPTERA is threefold. At the physical/link-layer level, it brings satellite FSO transmission beyond intensity-modulation direct-detection point-to-point link demonstrations, by combining coherent transmission, WDM (up to 10 wavelengths), aperture diversity employing space-time block codes and digital beam combining, probabilistically shaped dual-polarization quadrature amplitude modulation (QAM), adaptive DSP/forward error correction (FEC) with time and carrier recovery, interleaving/automatic repeat request/erasure recovery, link-budget-aware power control and transparent multi-hop forwarding without optical-electrical-optical regeneration (at least three hops). This enables the project to target up to ≥ 1 Tb/s per wavelength, fine rate granularity, and post-FEC bit error rate $\leq 10^{-15}$ under optically impaired conditions. At the network layer, OPTERA extends optical satellite networking beyond static or RF architectures by developing optical-aware routing, handover and telemetry-driven SDN mechanisms for dynamic LEO scenarios. These mechanisms exploit orbital predictability, visibility windows, PAT constraints, site diversity and time-varying link quality to reduce unnecessary route changes, optical-link reconfigurations and service interruptions. The key performance indicators (KPIs) include packet delivery ratio above 99% and $\geq 90\%$ under failures or overload, routing convergence ≤ 1 s, handover failure rate $\leq 1\%$, decision latency ≤ 50 ms, more than 40% header-overhead reduction compared with SRv6, path enforcement time below 5 ms and multi-agent coordination time below 500 ms. The third element is validation. The lab testbed combines optical subsystems, network nodes, and FSO impairment emulation using both COTS and proprietary hardware: digital-to-analog converter/analog-to-digital converter, C-band lasers, dual polarization in-phase/quadrature modulation, high-power amplification, coherent reception, programmable filtering, symbol rates up to 40 GBd, 64QAM, and up to 10 WDM channels. The final demo, targeting 2.5 Tb/s, runs UC1 and UC2 over at least two optically impaired links, with routing recovery ≤ 2 s, handover success $\geq 90\%$, service interruption ≤ 1 s, and $\geq 20\%$ improvement over static operation.

Methodology and research planning

OPTERA adopts an objective-oriented methodology structured around five work packages (WPs), thirteen task (Tx.y) objectives, quantitative KPIs, five milestones (MSs) and four deliverables (Fig 2). WP1 covers management, dissemination and exploitation, coordinating scientific, financial and ethical aspects and promoting results through a website, publications, public reports, a workshop, open repositories, a Python suite on GitHub, and follow-up opportunities. WP2 defines scenarios, requirements and architecture; its completion marks MS1 "System and network architecture and requirements defined". WP3 develops system models and design: T3.1 provides channel models for fiber, satellite-to-ground and ground-to-satellite FSO, and inter-satellite optical links; T3.2 designs coherent optical transmission, link budgets, WDM operation, adaptive power control and transparent forwarding; T3.3 develops DSP and coding. WP3 feeds networking and experimental validation and its completion marks MS2 "System model and design concluded". WP4 develops time-aware routing (T4.1), optical handover/link switching, resource allocation (T4.2) and telemetry-driven source routing (T4.3), addressing LEO dynamics, optical-terminal constraints, visibility windows, site diversity, adaptive control, and AI-assisted coordination; it leads to MS4 "Final programmable networking solution for space-ground systems". WP5 provides experimental validation: T5.1 develops the testbed, T5.2 validates the coherent optical satellite link and marks MS3 "Point-to-point optical link experimentally validated", and T5.3 demonstrates the complete space-ground optical network for the two UCs. WP5, together with WP1, culminates in MS5 "Network experimental demonstration completed and end of the project", and achievement of OPTERA's global objective. Deliverables include two annual and a final public report and an open-source simulation platform.

The methodology is interdisciplinary, combining optical systems, FSO propagation, satellite communications, coherent transmission, DSP, coding, optical hardware, routing, handover, SDN, AI-assisted adaptation and experimental photonic testbeds. CNR contributes to optical systems design, FSO propagation modelling, networking aspects, and coordination; SSSA contributes to coherent optical transmission, optical networking, and experimental validation; POLITO contributes to satellite communications, routing, handover, and network management; UNIVPM focuses on DSP, coding, simulation tools, dissemination, and exploitation. Experimental

validation will be carried out at SSSA, in an established photonics and optical-networking laboratory. Thanks to partners complementarity, OPTERA bridges physical-layer design, network-layer control, and laboratory implementation. Resource allocation, timing, objectives, activities, and milestones are shown in Fig 2.

Risk management is embedded in the methodology. Main risks concern coordination, budget deviations, leadership continuity, equipment procurement, control-loop instability, personnel recruitment, high optical powers involved. Mitigations include technical reviews, budget monitoring, distributed leadership, early procurement, adaptive thresholds and progressive validation, flexible personnel allocation and activities prioritization, lab procedures, and fallback emulation strategies.

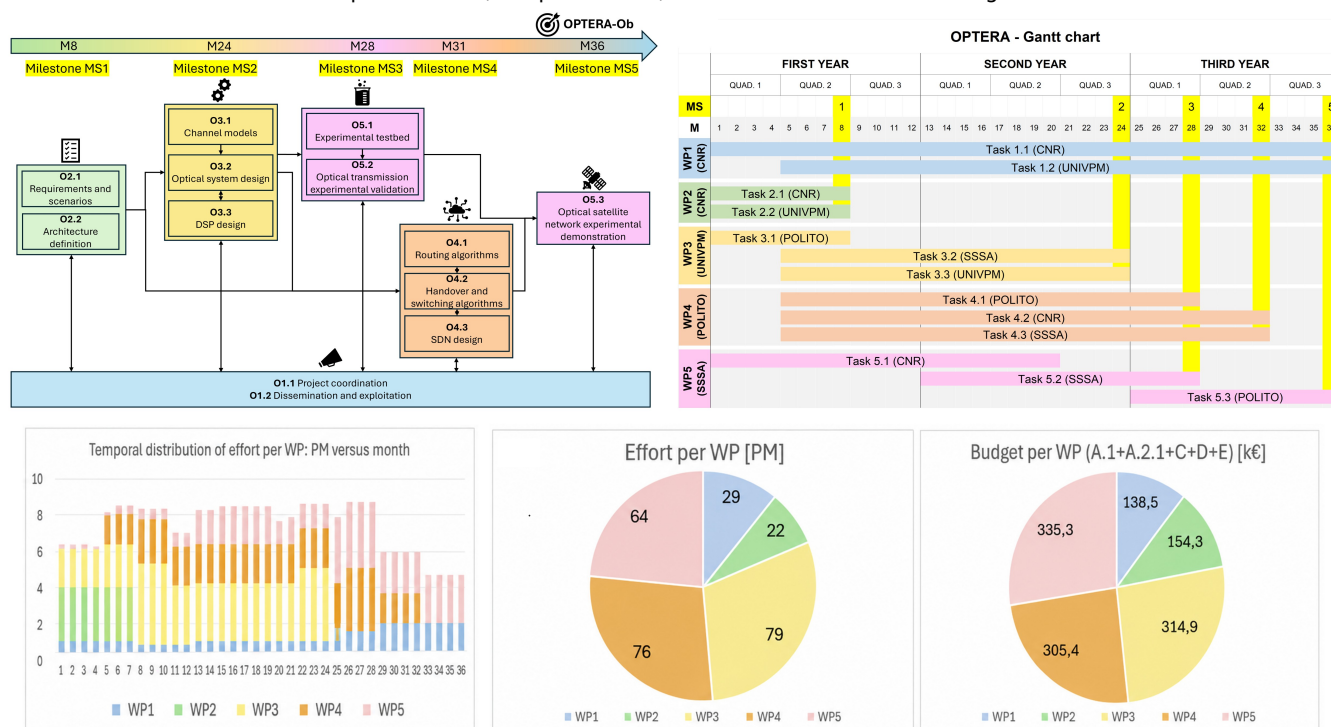


Fig 2: PERT, GANTT, activities, milestones, objectives, timing, and resource allocation

2. CURRICULA VITAE AND TRACK RECORD of the PI, the Associated PIs and Substitute Principal Investigator,
highlighting their expertise, scientific experience, and, where applicable, coordination skills.

PI

- Researcher unique identifier: ORCID Id

0000-0002-3571-202X

- URL for web site:

<https://scholar.google.it/citations?user=bwfn1sAAAAJ&hl=it>

Current and Previous positions

1. Since December 2022, I am a permanent researcher (III level) at CNR-IEIT, Pisa (IT).
2. From October 2018 to November 2022, I was a postdoctoral researcher "assegnista di ricerca" at Tecip Institute, Scuola Superiore Sant'Anna, Pisa (IT).
3. From October 2015 to September 2018, I was PhD student in "Emerging Digital Technologies, Photonic Technologies" at Tecip Institute, Scuola Superiore Sant'Anna, Pisa (IT). Advisor: M. Secondini. Thesis title: "Nonlinear frequency-division multiplexing: theoretical aspects, numerical algorithms, and experimental demonstration". Final mark: with honors.
4. Scientific qualification as associate professor in the Italian higher education system for 09/F2 - Telecommunications "Abilitazione scientifica nazionale alle funzioni di professore universitario di seconda fascia nel settore concorsuale 09/F2 TELECOMUNICAZIONI" obtained on March, 2025

Prizes and awards

1. GTTI PhD Award 2020 -awarded by Associazione Gruppo Telecomunicazioni e Tecnologie dell'Informazione (GTTI) for PhD Theses in the field of Communication Technologies

2. Best Doctoral Thesis Award in Applied Photonics 2019 – awarded by the IEEE Photonics Society
Italian Chapter

3. Nominated for the “Premio Giovanni Spitali - 2020” – awarded by Scuola Superiore Sant'Anna to the best PhD thesis

4. Best Student paper Award (2nd prize) – sponsored by the Electromagnetics Academy and awarded by the Subcommittee 3 of the PIERS Conference held in Toyama, Japan, in August 2018. Paper title: “Improved detection strategies for nonlinear frequency division multiplexing”.

5. Best Contributed Paper – awarded by The Rank Prize Funds, Optoelectronics Committee, at the Symposium on Challenges to Achieving Capacity in Nonlinear Optical Networks held in Grasmere Cumbria (UK) in June 2018.

Visiting academic positions	1. November 2017-May 2018. Visiting researcher at the Aston Institute of Photonic Technologies (UK) working with Prof. S.K. Turitsyn and his group on dual polarization nonlinear Fourier transform (NFT) based communication systems 2. May 2018. Visiting researcher at the Photonics Department, Technical University of Denmark (DK), working with Prof. D. Zibar and his group at the first experimental demonstration of NFT-based communication system with modulation on both polarizations and nonlinear spectra.
Teaching activities and PhD supervision	1. Tutor of the PhD student Dario Cellini, PhD in Photonic Technologies, Scuola Superiore Sant'Anna, supervisor M. Secondini, since 2023 2. Contract for Teaching Assistance Activities in L8 a.y. 2023/2024, L8 and LM31 a.y. 2022/2023, LM31 a.y. 2020/2021 at Universitas Mercatorum 3. Webinars for Industrial PhD "Big data e intelligenza artificiale" Universitas Mercatorum (3 webinars in a.y. 2023/2024 and two webinars in a.y. 2022/2023)
National and international collaborations and networks that contribute to enhancing the scientific quality and feasibility of the proposal	I have a strong international profile and an established reputation in optical communications. My network demonstrates both my scientific standing and my ability to successfully lead the proposed project. 1. Scientific recognition and community leadership I am well embedded in the international optical communications community, as demonstrated by my service on the Technical Program Committees of leading conferences, including SPPCOM and OFC. My recognition is further evidenced by numerous invited talks at major international conferences and workshops (SPPCOM 2025, ICTON 2025, ONDM 2025, IPC 2024, PSC 2024, ECOC 2023, OFC 2023, ISWCS 2021, SCACON 2018), a keynote lecture (MSCT 2026), and an invited industrial talk (Huawei 2025). I have also contributed to shaping the research agenda by organizing international workshops and panels (at ONDM 2025, ONDM 2026, OFC 2026, OFC 2027). 2. Participation and leadership in international projects I have extensive experience in international research projects, compared to peers at a similar career stage, including: (i) HORIZON-RIA ALLEGRO 101092766, where I was PI for CNR, third party of CNIT (23k€ for CNR); (ii) bilateral Italy-Israel R&D Cooperation Program MAE PERSUADER, where I was PI for CNR (20k€ for CNR); (iii) PNRR-RESTART, funded by the EU, where I was contact point for CNR for RIGOLETTO; (iv) Convenzione Operativa CNR-IEIIT and Tecip, Scuola Superiore Sant'Anna to foster research in optical fiber communications, I am PI for CNR (15k€ yearly for CNR in 2024-2025 and in 2025-2026); (v) participation in multiple industrial collaborations with Huawei (4 from 2020) and FiberHome (one in 2018-2019) 3. International collaborations driven by shared research interests, demonstrated by joint publications with Technical University of Denmark, Aston University, Technical University Eindhoven, Universitat Politècnica de Catalunya, Università di Firenze, Università di Pisa, Scuola Superiore Sant'Anna, CNIT

Other work experience (e.g. consultancy if any)	1.Universitas Mercatorum - Contract for Teaching Assistance Activities in L8 a.y. 2023/204, L8 and LM31 a.y. 2022/2023, LM31 a.y. 2020/2021 2.Universitas Mercatorum - 2 webinars for Industrial PhD in “Big Data ed Intelligenza Artificiale - XXXVIII Ciclo”, A.A. 2022/2023. 2.Universitas Mercatorum - 3 weminars for PhD in “Big Data ed Intelligenza Artificiale”, A.A. 2023/2024.
- Administrative role and position responsibility	1. Postdoctoral Researcher Representative (rappresentate degli assegnisti). From June 2022 till December 2022, I was representative of postdoc researchers in the council of the institute of telecommunications, computer engineering, and photonics (Tecip) of Scuola Superiore Sant’Anna. The role, of 3 years duration, concluded when I joined CNR-IEIIT. 2. I was contact point for all CNR units for RIGOLETTO project, a PNRR-RESTART project. 3. I was PI for CNR-IEIIT unit in PERSUADER project, an Italy-Israel R&D Cooperation Program MAE 2 years with CNIT, CyberRydge, CNR-IEIIT and CamGraphics (20k€ for CNR-IEIIT), 2024-2026. 4. I was PI for CNR unit as CNIT third party in ALLEGRO project, an HORIZON-RIA ALLEGRO project (23k€ for CNR), 2023-2026. 5. I am PI for CNR-IEIIT unit for the "Convenzione operativa CNR-IEIIT and Tecip, Scuola Superiore Sant’Anna" since December 2025 and previously from June 2024 till May 2025 (15k€ each for CNR-IEIIT).
- Scientific organisations/Coordination of academic activities	1.Workshop organization - ONDM 2025 "Securing the Light: Challenges and Innovations in Optical Fiber Security" 2.Workshop organization - ONDM 2026 "Advances in Network Security" 3. Panel organization - OFC 2026 "Probabilistic Amplitude Shaping after 10 Years: Where Are we Now, and What is to Come?" 4. Workshop organization - OFC 2027 "IM-DD vs Coherent Lite for Intra- and Inter-Building Data Center Interconnects"
Editorial activity	1. I serve as reviewer for several international journals as the IEEE Journal of Lightwave Technology, Optics Express, Optics Letter, and MDPI journals 2. I am a member of the technical program committee (TPC) of Optical Fiber Communication (OFC) Conference for S2 Transmission Subsystems since 2025, for which, beyond proposing invited and tutorial speakers, reviewing papers, and chairing technical sessions, I organized one panel for OFC 2026 and I am organizing a workshop for OFC 2027. 3. I was a member of the technical program committee (TPC) of Signal Processing in Photonic Communications (SPPCom - Advanced Photonics Congress) in 2022 and 2023.
Membership of scientific societies	1. I am member of Institute of Electrical and Electronics Engineers (IEEE), since 2015 2. I am affiliate researcher at Tecip Institute, Scuola Superiore Sant'Anna, since 2023. 3. I am member of Consorzio Nazionale Interuniversitario delle Telecomunicazioni (CNIT), since 2015.
Funding (current and past)	

Anno	Project title	Person months	Funding organisation	Role
2024	PERSUADER	7	MAECI Italy-Israel R&D cooperation program 2 years	unit PI; WP leader

2023	ALLEGRO	2.1	EU HORIZON-RIA, third party CNIT	unit PI; participant
2023	PNRR- RESTART RIGOLETTO	8.6	european union, PNRR	contant point for CNR in RIGOLETTO; participant

Significant career breaks	none
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Associated PIs

1. SECONDINI Marco

- Researcher unique identifier: ORCID Id	0000-0001-5883-5850
- URL for web site:	https://www.santannapisa.it/it/marco-secondini
Current and Previous positions	• 2022-pres: Associate Professor, Institute of Communication, Information, and Perception Technology (TeCIP), SSSA, Pisa, Italy. • 2019-2021: Tenure-Track Assistant Professor (RTDb), TeCIP Institute, SSSA, Pisa, Italy. • 2007-2018: Assistant Professor, TeCIP Institute, SSSA, Pisa, Italy. • 2005-2006: Research Fellowship (Assegno di Ricerca), TeCIP Institute, SSSA, Pisa, Italy. • 2002-2005: PhD student, TeCIP Institute, SSSA, Pisa, Italy. • 2001: Laboratory Technician, R&D, Qplus Network Italia, Roma, Italy. • 2000-2001: System Engineer, Space & Defense Area, D'Appolonia S.p.A., Roma, Italy.
Prizes and awards	• Top Scored Paper Award (first author and speaker) at International Conference on Optical Network Design and Modeling (ONDM 2015), Pisa, Italy • Top Reviewer Award, Journal of Lightwave Technology, 2016 • Best Student paper Award, 2nd prize (co-author and student supervisor) at PIERS 2018 Conference, Toyama, Japan • GTTI PhD Award 2020 (as thesis supervisor) awarded by Associazione Gruppo Telecomunicazioni e Tecnologie dell'Informazione (GTTI) for PhD Theses in the field of Communication Technologies • Best Doctoral Thesis Award in Applied Photonics 2019 (as thesis supervisor) awarded by IEEE Photonics Society Italian Chapter • Invited speaker to major international conferences in optical communications, including Optical Fiber Communication Conference (2016, 2020), European Conference on Optical Communication (2017, 2021), Asia Communications and Photonics Conference (2015)
Visiting academic positions	• Visiting Faculty Research Assistant, Photonics Laboratory, University of Maryland Baltimore County, Baltimore MD, USA, 2005.
Teaching activities and PhD supervision	COURSES • Inform. Theory & Applications, 30h, MSc, 2023-pres • Inform. Theory for Opt. Commun., 30h, PhD, 2023-pres • Fundam. of Opt. Commun., 30h, PhD, 2008-pres • Stochastic Processes & Queue Theory, lect. & exerc., 7h, MSc, 2014-2018 • Performance Evaluation of Comm. Syst., 24h, MSc, 2009-2014 • Introd. to Markov Chain & Multicanonical Monte Carlo Simul., 20h, PhD, 2007-2008 PHD SUPERV. • D. Cellini, Advanced Signal Processing Techniques for Opt. Commun., ongoing • E. Parente, Continuous-Variable Quantum Key Distrib., 2025

- M. R. Camara, Form Classical to Quantum Opt. Commun., 2019
- S. Civelli, Nonlinear Frequency-Division Multiplexing for Opt. Fiber Commun., 2019
- D. Marsella, Channel Modeling and Nonlinearity Mitigation Techniques in Coherent Opt. Systems, 2014
- L. Gerardi, Advanced Techniques for Performance Evaluation in Opt. Commun., 2009
- M. Frezzini, Analytical Performance Evaluation of Innovative Modulation Formats for Photonic Commun. Systems, 2009

National and international collaborations and networks that contribute to enhancing the scientific quality and feasibility of the proposal	• Joint research and co-authored publications with leading academic institutions: Eindhoven Univ of Technol (TuE, ICT Lab), Netherlands; Chalmers Univ of Technol (Fiber-Optic Communications Research Center, FORCE), Sweden; Aston Univ (Nonlinear Photonics and Fibre Lasers Group), UK; Techn Univ of Denmark (DTU, Signal Processing Group); Univ Coll London (UCL, Optical Networks Group), UK; Karlsruhe Inst of Technol (KIT, Communications Engineering Group), Germany; Univ. of Milan (Quantum Optics and Information Group); Univ of Parma (SPADiC Lab). • Collaborations with major industrial institutions: - Ericsson (Pisa Research Center): joint research and publications on fiber-optic systems, collaborative research projects, field trials with international telecom operators; - Huawei Technologies (China): joint research and publications on optical communications and advanced DSP techniques, collaborative research projects, advisory boards. • Exchange of visiting PhD students: Chalmers, TuE, DTU, Aston, KIT. • Teaching, thesis supervision, student exchanges within MAPNET and PIXNET Erasmus Mundus joint Master Programs: Aston Univ, Osaka Univ, Techn Univ Berlin, Eindhoven Univ of Techn • External examiner/reviewer/opponent for PhD candidates at Chalmers, KTH, Univ Toronto, Eindhoven Techn Univ, Denmark Techn Univ, Univ College London, Inst Polyt Paris, Univ Edinburgh, Politec Turin, Univ Parma • Joint master program and thesis supervision with University of Pisa • Invited seminars: Chalmers, Univ Coll London, Techn Univ Munich, European School of Information Theory, Ericsson, Huawei, Telefónica
Other work experience (e.g. consultancy if any)	• 2001-2004: Collaboration grants, Photonics Networks National Laboratory, CNIT, Pisa, Italy
- Administrative role and position responsibility	• Member of the Advisory Board (Collegio Docenti) of the International Doctoral School in "Emerging Digital Technologies", SSSA, academic years: all from 2017/2018 to present. • Member of the Selection Board for the admission to the International Doctoral School in "Emerging Digital Technologies", SSSA, academic years: 2010/2011; 2015/2016; 2018/2019; 2020/2021; 2022/2023; 2026/2027. • Chair of the Degree Examination Committee (final degree examinations), SSSA, sessions: March 2025, June 2025, December 2025. • Member of the Selection Board for many post-doc and RTDA positions.
- Scientific organisations/Coordination of academic activities	• TPC Member: Optical Communication Conference (OFC) 2021, Subcommittee S5: Digital transmission systems • TPC Member: Asia Communications and Photonics Conference (APC) 2019, Chengdu, China, 2019 • TPC Member: European Conference on Optical Communication (ECOC) 2019, SC5 – Theory of Optical Communications, Dublin, Ireland, 2019 • Co-organizer: Rank Prize Funds Symposium (http://www.rankprize.org/index.php/symposia) on "Challenges to Achieving Capacity in Nonlinear Optical Networks, Grasmere, UK, 2018 • Co-organizer: Workshop "Nonlinear Fourier Transform in Optical Communications" (https://sites.google.com/site/nocworkshop/home), Florence, Italy, 2018 • Chair of the Technical Program Committee (TPC): Tyrrhenian International Workshop on Digital Communications (TIWDC) 2015, Fiber Nonlinearities in Coherent Optical Communications", Florence, Italy, http://tiwdc2015.sssup.it (sponsored by IEEE) • TPC Member: IEEE Global Communication Conference (Globecom), years 2010, 2011, 2015, http://globecom2015.ieee-globecom.org/ • TPC Member: IEEE International Conference on Communications (ICC), years 2010, 2011, 2012, 2013, 2014, 2015, 2016, http://icc2016.ieee-icc.org/ • TPC Member: OSA Topical Meeting on Photonics in Switching (PS 2014), San Diego, CA, USA,

<https://www.osapublishing.org/conference.cfm?meetingid=35&yr=2014>

- PI: MIUR FIRB project "COherent Terabit Optical NETworks (COTONE)", SSSA + CNIT + UNIPR, 2012-2015
- Local PI: Min. Dif. PNRM Project "QUAntum SAfe netwoRk (QUASAR)", UNIROMA1 + SSSA + CNIT + UNINA + UNIAQ
- Local coord.: H2020-Marie -Skłodowska-Curie action COFUND Project "Multiply", 2015 led by Aston Univ
- Task leader: PNRR RESTART project "Rigoletto", Task 3.5 "Optical communication theory and security", SSSA + CNR + POLITO + UNINA + Open Fiber, 2023-2026
- Leader of Spoke 7 activity at SSSA unit, PNRR Project "National Quantum Science and Technology Institute (NQSTI)", 2023-2025
- Session chair: OFC 2021, ECOC 2019, GLOBECOM 2010

Editorial activity	• Guest Editor, IEEE Journal of Selected Areas in Communications (impact factor: 17.2), Special Issue "Next-generation optical communications and networking", 2025. • Associate Editor, IEEE Transaction on Communications (impact factor: 8.3), from February 2018 to March 2023. • Guest Editor, Entropy (impact factor: 2.524), Special Issue "Information Theory of Optical Fibers", 2020.
Membership of scientific societies	• Senior Member, Institute of Electrical and Electronics Engineers (IEEE) since 2017 (Member since 2015).

	Anno	Project title	Person months	Funding organisation	Role
Funding (current and past)	2025	QUAntum SAfe netwoRk (QUASAR) - Fase II	4	Italian Ministry of Defense - PNRM Project	Local PI
	2025	Research on fiber nonlinear compensation algorithm and constellation shaping scheme for Raman amplification link	4	Huawei Technologies	PI
	2022	QUAntum SAfe netwoRk (QUASAR) - Fase I	3	Italian Ministry of Defense - PNRM Project	Local PI
	2021	Nonlinear constellation shaping and digital backpropagation for fiber nonlinearity mitigation	4	Huawei Technologies	PI
Advance Fiber					

2019	Nonlinear Compensation Technology Cooperation Project	4	Huawei Technologies	PI
2017	Nonlinear spectrum modulation (NOSTRUM)	3	Cassa di Risparmio di Firenze	Local PI
2015	COherent Terabit Optical NETworks (COTONE)	42	Italian Ministry for Education University and Research - FIRB Project Futuro in Ricerca	PI
Significant career breaks				
None				

2. BATTAGLIONI Massimo

- Researcher unique identifier: ORCID Id	0000-0002-8539-4007
- URL for web site:	https://scholar.google.it/citations?user=jrfv5IsAAAAJ&hl=en
Current and Previous positions	Since July 2024: Tenure-Track Researcher (RTT) in Telecommunications at the Department of Information Engineering (DII) of UNIVPM. Research topics: communication reliability and security, with a focus on coding theory and its applications to wireless and satellite communications. From April 2020 to June 2024: Research Fellow (Assegnista di Ricerca) at UNIVPM, working on projects related to cyber risk assessment, including the design of algorithms and protocols for the analysis and mitigation of cyber threats. From January 2019 to April 2020, Research Fellow at UNIVPM, focusing on the design and analysis of coding schemes for reliable and secure communications, with particular emphasis on LDPC codes and their applications. From November 2015 to October 2018: PhD in Information Engineering (curriculum: Biomedical, Electronics and Telecommunications Engineering) at UNIVPM. Final grade (March 2019): Outstanding, with a thesis entitled "Design and Analysis of Spatially Coupled LDPC Convolutional Codes", advisor: Prof. Giovanni Cancellieri.
Prizes and awards	1. GTTI PhD Award for the best PhD theses in the field of Communication Technologies (2019), awarded by the National Telecommunications and Information Theory Group (GTTI). Thesis title: Design and analysis of spatially coupled LDPC convolutional codes 2. IEEE Communications Letters Exemplary Reviewer Award in 2018 3. IEEE Communications Letters Exemplary Reviewer Award in 2019 4. IEEE Communications Letters Exemplary Reviewer Award in 2019 5. IEEE Transactions on Communications Exemplary Reviewer Award in 2019 6. IEEE Communications Letters Exemplary Editor Award in 2021 7. IEEE Communications Letters Exemplary Editor Award in 2025

Visiting academic positions	1. From April to July 2017: visiting PhD student at Lund University (Department of Electrical and Information Technology, LTH), working on coding theory. Supervisor: Prof. Lentmaier. 2. From May to August 2018: visiting PhD student at New Mexico State University (Klipsch School of Electrical and Computer Engineering), focusing on the design and analysis of spatially coupled codes. Supervisor: Prof. Mitchell. 3. In September 2018: visiting PhD student at University College Dublin (School of Electrical and Electronic Engineering), working on coding techniques for reliable relay communications. Supervisor: Prof. Flanagan.
Teaching activities and PhD supervision	Current teaching (UNIVPM): 1. Information Theory & Coding (MSc, 72 h, A.Y. 2023–2026) 2. Communication Technologies for Environmentally Sustainable Systems (BSc, 48 h, A.Y. 2024–2026) 3. Communication Networks (BSc, 36 h, A.Y. 2025–2026) Past teaching: 1. Digital Communications (MSc, 72 h, A.Y. 2020–2023), UNIVPM 2. Telecommunications Network Security (MSc, 48 h, A.Y. 2020–2024), eCampus 3. Cybersecurity and Blockchain (MSc, 20 h, A.Y. 2022–2024), UNIMC PhD supervision: 1. Supervisor of E. Carnevali, since Nov. 2025 - “Reliability of wireless communications in indoor environments for automation and smart logistics” 2. Co-supervisor of A. Baldelli, since Nov. 2024 - “Design and cryptanalysis of post-quantum code-based cryptographic schemes and protocols” 3. Co-supervisor of R. Giuliani, since Nov. 2023 - “Design and validation of new techniques based on coding and cryptography for reliability and security of data and networks”
National and international collaborations and networks that contribute to enhancing the scientific quality and feasibility of the proposal	At the international level, on topics related to communications reliability, I have collaborated and collaborate with institutions including Lund University, University of Bergen, University of Zurich, Florida Atlantic University, New Mexico State University, University of Notre Dame, Technical University of Munich, Karlsruhe Institute of Technology, and University of Arizona. These collaborations have resulted in joint research activities and publications. At the national level, some of my collaborations are with Politecnico di Torino, Università di Bologna, Università di Trieste, and Università dell'Aquila. I have also been involved in several research projects, including: - ECONET - Efficient and reliable communication networks, funded by the European Union under Next Generation EU / NRRP; - QSAFEIT - Quantum-safe cryptographic tools for the protection of national data and information technology assets, funded by Ministry of University and Research (MUR) under the Italian Fund for Applied Sciences (FISA); - AQUSDIT - Advanced and Quantum-safe Solutions for Digital Identity and digital Tracing, funded by the European Union under Next Generation EU / NRRP; - MASSCON - Multiple access sequences for satellite constellations without centralised management, funded by ESA/ESTEC; - OFF-A - Offline Antenna Correlator for EnVision, funded by ESA/ESOC; - Telemetry randomizer for high data rates, funded by ESA/ESOC; - CYBERAMA - Cyber Risk Assessment Models and Algorithms, funded by Fondazione Cariverona; - CTE SQUARE, funded by the Italian Ministry of Enterprises and Made in Italy. In the context of satellite communication systems, I also maintain collaborations with industrial partners such as Arpsoft, Tekne, and Thales Alenia Space, as well as with the European Space Agency and the German Aerospace Center.
Other work experience (e.g. consultancy if any)	I performed consultancy activities through the National Inter-university Consortium for Telecommunications (CNIT) on error-correcting codes for space communications, including studies on non-conventional coding techniques (November 2018) and on the impact of tail sequences on satellite communication system performance (November 2022 - January 2023). I have also been involved in projects funded by the European Space Agency, including "Telemetry randomizer for high data rates" (2019–2020) and "Offline Antenna Correlator for EnVision (OFF-

A)" (2022–2024), the latter as consultant, as well as in the "Multiple access sequences for satellite constellations without centralised management (MASSCON)" project (2023–present) as WP leader, in collaboration with industrial partners such as Thales Alenia Space, Arpsoft, and Zelinda.

- Administrative role and position responsibility	1. Principal Investigator of the project "Efficient and Reliable Communication Networks" (ECONET) (2024–2025), funded under the Italian NRRP (NextGenerationEU) with a total budget of €497,209.25, involving academic and industrial partners.
	2. Work Package Leader in the European Space Agency project MASSCON (2023–present), funded by ESA/ESTEC, focused on multiple-access techniques for satellite constellations.
	3. Member of the Action Plan Implementation Committee for the Human Resources Strategy for Researchers, responsible for overseeing and coordinating the execution of strategic HR actions, monitoring progress, and ensuring alignment with institutional and European research standards, since 2026.
- Scientific organisations/Coordination of academic activities	1. Local Arrangement Co-Chair of the 2025 IEEE European School of Information Theory (ESIT)
	2. Co-chair of the special session "Recent Advances in Coding Theory and its Applications" at the AEIT 2019 International Annual Conference
	3. Chair of sessions at international conferences, including AEIT 2019, IEEE Information Theory Workshop 2020, and IEEE Global Communications Conference 2023.
Editorial activity	1. Associate Editor for IEEE Communications Letters (October 2020 - February 2026)
	2. Reviewer for several international journals including, among many others IEEE Communications Letters, IEEE Transactions on Communications, IEEE Transactions on Information Theory, and IEEE Journal on Selected Areas in Communications.
	3. TPC member or committee member for several conferences and symposia, including AEIT International Annual Conference, the Symposium on New Trends in Satellite and Space Communications and Navigation @ SoftCOM 2025, BCCA 2025, and IEEE ISIT 2026.
	4. Recipient of the IEEE Communications Letters Exemplary Reviewer Award in 2018, 2019, and 2020, the IEEE Transactions on Communications Exemplary Reviewer Award in 2019, and the IEEE Communications Letters Exemplary Editor Award in 2021 and 2025.
Membership of scientific societies	1. Member of the Institute of Electrical and Electronics Engineers (IEEE)
	2. Member of the IEEE Communications Society (ComSoc)
	3. Member of the the IEEE Information Theory Society (ITSoc)
	4. Member of the National Telecommunications and Information Theory Group (GTTI)
	5. Affiliated with the National Inter-university Consortium for Telecommunications (CNIT).

Anno	Project title	Person months	Funding organisation	Role
2024	Efficient and Reliable	6.22	European Union - Next Generation EU under the Italian National	Scientific Coordinator

	Communication Networks (ECONET)		Recovery and Resilience Plan (NRRP)	
Funding (current and past)				
	2024	Advanced and Quantum-safe Solutions for Digital Identity and digital Tracing (AQuSDIT)	4.8	European Union - Next Generation EU under the Italian National Recovery and Resilience Plan (NRRP) Team member
	2024	Quantum-safe cryptographic tools for the protection of national data and information technology assets (QSAFEIT)	0.22	Ministry of University and Research (MUR) under the Italian Fund for Applied Sciences (FISA) Team member
Significant career breaks	Nothing to declare			
3. GARELLO Roberto				
- Researcher unique identifier: ORCID Id	0000-0003-0292-4648			
- URL for web site:	https://www.polito.it/personale?p=roberto.garello			
Current and Previous positions	2001 - present: Associate Professor - Politecnico di Torino 1998 - 2001: Associate Professor - Università di Ancona			
Prizes and awards	Best Paper Award for - 9th TT&C International Workshop, 2022 - The Third International Conference on Advances in Cognitive Radio - COCORA 2013 - The Third International Conference on Communication Theory, Reliability, and Quality of Service - CTRQ 2010 - SIP award for his PhD thesis - Fondazione Ugo Bordoni award for his MoS thesis			
Visiting academic positions	- California State University, Los Angeles, Adjunct Professor, 2017 - MIT and ETH, visiting PhD, 1993			
Teaching activities and PhD supervision	Responsible (titolare) for more than 60 university courses as course coordinator, first at the University of Ancona and later at Politecnico di Torino.			
	Lecturer for other 20 courses at Politecnico di Torino.			
	Responsible for Communication Technologies at UniNettuno, Rome, from 2013 to 2017.			
	Lecturer for two courses in the Master of Science in Electrical Engineering at California State University, Los Angeles.			
	Advisor of more than 200 Master of Science theses, including 100 carried out in collaboration with companies.			
	Coordinator of the Master of Science degree in Communications Engineering at Politecnico di			

Torino from 2006 to 2008.

Coordinator of the distance-learning degree program in Telecommunications Engineering at Politecnico di Torino from 2003 to 2008.

PhD advisor of 13 doctoral students who successfully completed their PhD, with 3 additional PhD students currently under supervision.

National and international collaborations and networks that contribute to enhancing the scientific quality and feasibility of the proposal	Major international research collaborations resulting in joint publications (outside Italy): MIT, JPL, UCLA, ENST Bretagne, California State University Los Angeles, Centre Tecnològic de Telecommunicacions de Catalunya, Chalmers University of Technology, Comisión Nacional de Actividades Espaciales (CONAE), German Aerospace Center (DLR), Harbin Institute of Technology, Inria / INSA Lyon, Monash University, University of Auckland, University of the Bundeswehr Munich, University of Luxembourg, Surrey Space Centre, European Space Agency. Major national research collaborations resulting in joint publications (within Italy) Università di Padova, Università di Pisa, CNR, INRIM, ISMB / Links Foundation, Università di Bologna, Università di Torino, Università Politecnica delle Marche.
Other work experience (e.g. consultancy if any)	Marconi Communications, Genova, 1994-1997
- Administrative role and position responsibility	Member of the Board of the Department of Electronics and Telecommunications, Politecnico di Torino, since 2024. Member of the Board of the School of Master and Lifelong Learning, Politecnico di Torino, since 2025. Member of the Executive Committee of the School of Master and Lifelong Learning, Politecnico di Torino, since 2026. Member of CNIT Executive Board (Ancona unit) from 1999 to 2000 and of CNIT Scientific Board (Politecnico di Torino unit) from 2011 to 2013 Coordinator of the Master of Science degree in Communications Engineering at Politecnico di Torino from 2006 to 2008. Coordinator of the distance-learning degree program in Telecommunications Engineering at Politecnico di Torino from 2003 to 2008. Coordinator of internships for the Master of Science in Communications Engineering, Politecnico di Torino, since 2019.
- Scientific organisations/Coordination of academic activities	Since 2008, he has led his own research group, initially named Wi2P and now Astracom, which has hosted more than 15 PhD students as well as a similar number of postdoctoral researchers and research fellows. The Astracom group, active in space communication systems and covering physical layer, networking, and applications, is currently composed of two professors, three postdoctoral researchers, and four PhD students.
Editorial activity	Editor of Journal of Communications Software and Systems (JCOMSS), an international journal indexed in Scopus, since 2006. Guest Editor of the Special Issue "Channel Coding in Wireless Systems", Journal of Communications Software and Systems, with Marco Chiani and Franco Chiaraluce, 2006. Guest Editor of the Special Issue "Channel Coding Techniques for Next-Generation Communication Systems: Chances and Challenges", Electronics, with Gianluigi Liva and Francesca Vatta, 2021.
Membership of scientific societies	Member of the Institute of Electrical and Electronics Engineers (IEEE) since 1990; elevated to Senior Member in 2003. Associated Researcher at the Institute of Electronics, Computer and Telecommunication Engineering (IEIT), National Research Council (CNR), since March 2024.

	Anno	Project title	Person months	Funding organisation	Role
Funding (current and past)	2026	End-to-end demonstrator for 6G candidate waveforms (STARGATE)	3	European Space Agency	Politecnico di Torino Unit Leader
	2025	5G-NTN VLEO constellation for broadband D2D connectivity (FRIISAT)	3	European Space Agency	Politecnico di Torino Unit Leader
	2024	Handover, Data Routing And Radio Resource Management For Very Low Earth Orbit (Vleo) Broadband Constellations (HANDING OVER)	3	European Space Agency	Politecnico di Torino Unit Leader
	2023	Multiple Access Sequences for Satellite Constellations without Centralised Management	3	European Space Agency	CNIT Unit Leader
	2023	Physical Layer Simulator For 5G New Radio and OTFS	3	European Space Agency	Project Manager
	2022	On the use of mega-constellations services in space	3	European Space Agency	Politecnico di Torino Unit Leader
Significant career breaks	N/A				

PI and Associated PIs Scientific publications

n°	Publication	Name
1.	Civelli S, Forestieri E, Secondini M (2024). Sequence-Selection-Based Constellation Shaping for Nonlinear Channels. JOURNAL OF LIGHTWAVE TECHNOLOGY, vol. 42, p. 1031-1043, ISSN: 0733-8724, doi: 10.1109/JLT.2023.3332487	CIVELLI Stella
2.	Civelli S., Parente E., Forestieri E., Secondini M. (2023). On the Nonlinear Shaping Gain With Probabilistic Shaping and Carrier Phase Recovery. JOURNAL OF LIGHTWAVE TECHNOLOGY, vol. 41, p. 3046-3056, ISSN: 0733-8724, doi: 10.1109/JLT.2023.3241449	CIVELLI Stella
3.	Marco Secondini, Stella Civelli, Enrico Forestieri, Lareb Zar Khan (2022). New Lower Bounds on the Capacity of Optical Fiber Channels via Optimized Shaping and Detection. JOURNAL OF LIGHTWAVE TECHNOLOGY, vol. 40, p. 3197-3209, ISSN: 0733-8724, doi: 10.1109/JLT.2022.3148322	CIVELLI Stella
4.	Secondini, Marco, Agrell, Erik, Forestieri, Enrico, Marsella, Domenico, Camara, Menelaos Ralli (2019). Nonlinearity Mitigation in WDM Systems: Models, Strategies, and Achievable Rates. JOURNAL OF LIGHTWAVE TECHNOLOGY, vol. 37, p. 2270-2283, ISSN: 0733-8724, doi: 10.1109/JLT.2019.2901908	SECONDINI Marco
5.	SECONDINI, Marco, Foggi, Tommaso, FRESI, Francesco, MELONI, GIANLUCA, Cavaliere, Fabio, Colavolpe, Giulio, FORESTIERI, Enrico, POTI', LUCA, Sabella, Roberto, PRATI, Giancarlo (2015). Optical Time-Frequency Packing: Principles, Design, Implementation, and Experimental Demonstration. JOURNAL OF LIGHTWAVE TECHNOLOGY, vol. 33, p. 3558-3570, ISSN: 0733-8724, doi: 10.1109/JLT.2015.2443876	SECONDINI Marco
6.	Massimo Battaglioni, Kenneth Andrews, Rebecca Giuliani, Fabrizio Marinelli, Franco Chiaraluce, Marco Baldi (2025). Design and Analysis of the Tail Sequence for Short LDPC-Coded Space Communications. IEEE TRANSACTIONS ON AEROSPACE AND ELECTRONIC SYSTEMS, vol. 61, ISSN: 0018-9251, doi: 10.1109/TAES.2025.3599498	BATTAGLIONI Massimo
7.	Massimo Battaglioni, Alireza Tasdighi, Giovanni Cancellieri, Franco Chiaraluce, Marco Baldi (2018). Design and Analysis of Time-Invariant SC-LDPC Convolutional Codes with Small Constraint Length. IEEE TRANSACTIONS ON COMMUNICATIONS, vol. 66, p. 918-931, ISSN: 0090-6778, doi: 10.1109/TCOMM.2017.2774821	BATTAGLIONI Massimo
8.	Mingyi Wang, Yizhou Peng, Ruofei Ma, Gongliang Liu, Weixiao Meng, Carla Fabiana Chiasserini, Roberto Garelo (2026). Spatiotemporal-Attention Based Channel Prediction for UAV-RIS-Assisted LEO Satellite MIMO Communications. IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, vol. 25, p. 7252-7267, ISSN: 1536-1276, doi: 10.1109/TWC.2025.3630206	GARELLO Roberto
9.	Tuninato, Riccardo, Maiolini Capez, Gabriel, Mazzali, Nicolò, Garelo, Roberto (2025). 5G New Radio for Non-Terrestrial Networks: Analysis and Comparison of HARQ and RLC ARQ Performance Over Satellite Links. IEEE ACCESS, vol. 13, p. 75400-75415, ISSN: 2169-3536, doi: 10.1109/access.2025.3563983	GARELLO Roberto
10.	Capez, Gabriel Maiolini, Cáceres, Mauricio A., Armellin, Roberto, Bridges, Chris P., Fraire, Juan A., Frey, Stefan, Garelo, Roberto (2024). On the Use of Mega Constellation Services in Space: Integrating LEO Platforms into 6G Non-Terrestrial Networks. IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS, vol. 42, p. 1-15, ISSN: 0733-8716, doi: 10.1109/JSAC.2024.3459078	GARELLO Roberto

3. Main staff involved (professors/researchers for each research unit, in addition to the PI or associated PIs), highlighting the expected commitment.

List of the Research Units

Unit 1 - CIVELLI Stella

Personnel of the research unit

n°	Surname Name	Qualification	University/ Research Institution	e-mail address	Months/person expected
1.	CIVELLI Stella	Ricercatore	Consiglio Nazionale delle Ricerche	stella.civelli@cnr.it (adesione completata il 27/05/2026)	20,0
2.	TARABLE Alberto	Primo ricercatore	Consiglio Nazionale delle Ricerche	alberto.tarable@polito.it (adesione completata il 13/05/2026)	15,0

Information on the new contracts for personnel to be specifically recruited

Number of expected new contracts	Overall expected time commitment (months)
2	36

Unit 2 - SECONDINI Marco

Personnel of the research unit

n°	Surname Name	Qualification	University/ Research Institution	e-mail address	Months/person expected
1.	SECONDINI Marco	Professore Associato (L. 240/10)	Scuola Superiore di Studi Universitari e Perfezionamento Sant'Anna	m.secondini@santannapisa.it (adesione completata il 27/05/2026)	14,0
2.	FORESTIERI Enrico	Professore Ordinario	Scuola Superiore di Studi Universitari e Perfezionamento Sant'Anna	forestieri@sssup.it (adesione completata il 25/05/2026)	5,0

Information on the new contracts for personnel to be specifically recruited

Number of expected new contracts	Overall expected time commitment (months)
2	36

Unit 3 - BATTAGLIONI Massimo

Personnel of the research unit

n°	Surname Name	Qualification	University/ Research	e-mail address	Months/person
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			Institution		expected
1.	BATTAGLIONI Massimo	Ricercatore a t.d.-t.pieno (L. 79/2022)	Università Politecnica delle MARCHE	m.battaglioni@staff.univpm.it (adesione completata il 26/05/2026)	18,0

Information on the new contracts for personnel to be specifically recruited

Number of expected new contracts	Overall expected time commitment (months)
4	48

Unit 4 - GARELLO Roberto

Personnel of the research unit

n°	Surname Name	Qualification	University/ Research Institution	e-mail address	Months/person expected
1.	GARELLO Roberto	Professore Associato confermato	Politecnico di TORINO	roberto.garello@polito.it (adesione completata il 25/05/2026)	11,0
2.	VISINTIN Monica	Professore Associato confermato	Politecnico di TORINO	monica.visintin@polito.it (adesione completata il 26/05/2026)	10,0
3.	BENSO Nicolo'	Dottorando (data fine contratto: 31/10/2028)	Politecnico di TORINO	nicolo.bensof@gmail.com (adesione completata il 26/05/2026)	9,0

Information on the new contracts for personnel to be specifically recruited

Number of expected new contracts	Overall expected time commitment (months)
2	36

4. Time schedule of the research activities (GANTT CHART)

Milestone 1 System and network architecture and requirements defined

ACTIVITY	ASSIGNED TO	I year			II year			III year		
		QUAD. 1	QUAD. 2	QUAD. 3	QUAD. 1	QUAD. 2	QUAD. 3	QUAD. 1	QUAD. 2	QUAD. 3

Scenarios and system requirements (T2.1)	CIVELLI S SECONDINI M BATTAGLIONI M GARELLO R	X	X							
Reference architecture definition (T2.2)	CIVELLI S SECONDINI M BATTAGLIONI M GARELLO R	X	X							

Milestone 2 System model and design concluded

ACTIVITY	ASSIGNED TO	I year			II year			III year		
		QUAD. 1	QUAD. 2	QUAD. 3	QUAD. 1	QUAD. 2	QUAD. 3	QUAD. 1	QUAD. 2	QUAD. 3
Channel model (T3.1)	CIVELLI S SECONDINI M BATTAGLIONI M GARELLO R	X	X							
Optical system design (T3.2)	CIVELLI S SECONDINI M BATTAGLIONI M GARELLO R		X	X	X	X	X			
Digital signal processing (T3.3)	CIVELLI S SECONDINI M BATTAGLIONI M GARELLO R		X	X	X	X	X			

Milestone 3 Point-to-point optical link experimentally validated

ACTIVITY	ASSIGNED TO	I year			II year			III year		
		QUAD. 1	QUAD. 2	QUAD. 3	QUAD. 1	QUAD. 2	QUAD. 3	QUAD. 1	QUAD. 2	QUAD. 3
Development of a lab-scale testbed (T5.1)	CIVELLI S SECONDINI M BATTAGLIONI M	X	X	X	X	X				

Lab demonstration of coherent optical satellite link (T5.2)	CIVELLI S SECONDINI M BATTAGLIONI M GARELLO R				X	X	X	X		
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Milestone 4 Final programmable networking solution for space-ground systems

ACTIVITY	ASSIGNED TO	I year			II year			III year		
		QUAD. 1	QUAD. 2	QUAD. 3	QUAD. 1	QUAD. 2	QUAD. 3	QUAD. 1	QUAD. 2	QUAD. 3
Fast and robust time-aware routing algorithm design (T4.1)	CIVELLI S SECONDINI M BATTAGLIONI M GARELLO R		X	X	X	X	X	X		
Handover and link switching & resource allocation for optical links (T4.2)	CIVELLI S SECONDINI M BATTAGLIONI M GARELLO R		X	X	X	X	X	X	X	
Telemetry-driven segment routing framework (T4.3)	CIVELLI S SECONDINI M GARELLO R		X	X	X	X	X	X	X	

Milestone 5 Network experimental demonstration completed and end of the project

ACTIVITY	ASSIGNED TO	I year			II year			III year		
		QUAD. 1	QUAD. 2	QUAD. 3	QUAD. 1	QUAD. 2	QUAD. 3	QUAD. 1	QUAD. 2	QUAD. 3
Project Management (T1.1)	CIVELLI S SECONDINI M BATTAGLIONI M GARELLO R	X	X	X	X	X	X	X	X	X
Dissemination and exploitation (T1.2)	CIVELLI S SECONDINI M BATTAGLIONI M GARELLO R		X	X	X	X	X	X	X	X
	CIVELLI S									

Experimental demonstration of space-ground networks (T5.3)	SECONDINI M BATTAGLIONI M GARELLO R							X	X	X
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SECTION C - SCIENTIFIC PROPOSAL

1. SCIENTIFIC PROPOSAL

Detailed description of the project:

a. Clarity, relevance, and originality of the project objectives; positioning of the project with respect to the state of the art in the specific scientific area; soundness of the methodology adopted in relation to the objectives, degree of interdisciplinarity, and overall coherence of the project with respect to the specific contribution of the local units; identification of the main project risks and adequate mitigation strategies

List of OPTERA's acronyms

Acronym	Description	Acronym	Description	Acronym	Description
A2A	agent-to-agent	Gb/s	gigabits per second	PAT	pointing, acquisition, and tracking
ADC	analog-to-digital converter	GMI	generalized mutual information	PDR	packet delivery ratio
AI	artificial intelligence	GNN	graph neural network	PERT	program evaluation and review technique
ARQ	automatic repeat request	Gs/s	gigasample per second	PET	path enforcement time
ASI	Agenzia Spaziale Italiana	HPOA	high-power optical amplifier	PI	principal investigator
BER	bit error rate	IETF	Internet engineering task force	PM	person-month
CPR	carrier phase recovery	ISL	inter-satellite link	QAM	quadrature amplitude modulation
COTS	commercial off-the-shelf	ITU	International Telecommunication Union	RF	radio-frequency
DAC	digital-to-analog converter	KPI	key performance indicator	S2G	satellite-to-ground
dBm	decibel milliwatt	LCoS	liquid crystal on silicon	SDC	space data center
DP	dual polarization	LLM	large language model	SDG	sustainable development goal
DP-IQ	dual-polarization in-phase/quadrature	LEO	low-Earth orbit	SDN	software-defined networking
DP-QAM	dual-polarization quadrature amplitude modulation	MCP	model-context protocol	SNR	signal-to-noise ratio
DRL	deep reinforcement learning	MIMO	multiple-input and multiple-output	SRv6	Segment Routing over IPv6
DSP	digital signal processing	MS	milestone	SWaP	size, weight, and power
EO	Earth observation	NETCONF	Network Configuration Protocol	Tb/s	terabits per second
ESA	European Space Agency	NTN	non-terrestrial network	TN	terrestrial network
ESTOL	ESA specifications for terabit/sec optical links	OEO	optical-electrical-optical	TRL	technology readiness level
ETSI	European telecommunications standards institute	OIF	optical internetworking forum	UC	use case
FEC	forward error correction	ONF	open networking foundation	WDM	wavelength-division multiplexing
FSO	free-space optical	OPTERA	OPTical satellite multi-tERAbit communication network	WP	work package
G2S	ground-to-satellite	OPTERA-Ob	OPTERA global objective	YANG	Yet Another Next Generation
GBd	gigabaud	PAS	probabilistic amplitude shaping		

GLOBAL OPTERA OBJECTIVE

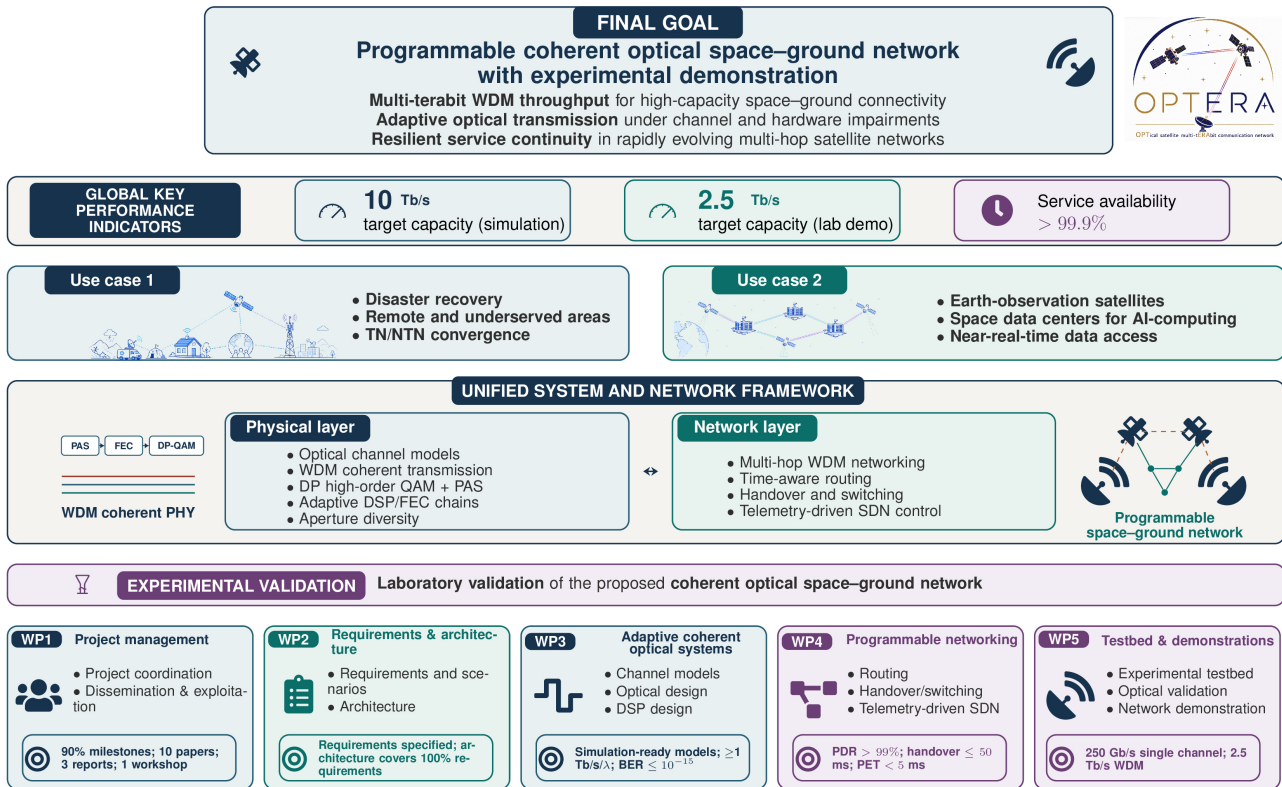


Fig a.1: OPTERA-Ob

Objectives

The global OPTical satellite multi-TERAbit communication network (OPTERA) objective (OPTERA-Ob) (Fig a.1) is to design and experimentally validate in a laboratory environment a scalable end-to-end programmable optical coherent space-ground network (TRL 4) achieving multi-terabit wavelength-division multiplexing (WDM) aggregate throughput, dynamic adaptation to realistic optical impairments, and resilient service continuity under rapidly evolving multi-hop satellite network conditions.

OPTERA targets two use cases (UCs): (UC1) resilient broadband/service continuity for disaster recovery, remote areas, and terrestrial/non-terrestrial networks (TN/NTNs) convergence (Fig a.2); (UC2) space data centers (SDCs) for AI computing and always-connected Earth observation (EO) satellites (Fig a.3).

OPTERA targets an optical next-generation space network that, compared with radio-frequency (RF) links, offers higher capacity and bandwidth, highly directional narrow beams, reduced interference risk, absence of spectrum regulations, and improved spectrum scalability. OPTERA exploits mature terrestrial optical technologies for satellite-to-ground (S2G), ground-to-satellite (G2S), and inter-satellite links (ISLs), while addressing the bottlenecks of existing optical links.

OPTERA integrates coherent WDM transmission, adaptive digital signal processing (DSP)/forward error correction (FEC), transparent optical forwarding, time-aware routing, handover, telemetry-driven software-defined networking (SDN) control, and AI-assisted adaptation, with laboratory validation of the proposed architecture.

Global OPTERA key performance indicators (KPIs): target system transmission capacity of 10 Tb/s (laboratory validation of 2.5 Tb/s); experimental demonstration of end-to-end service availability above 99.9% across the global coverage area, under dynamic multi-hop space-ground conditions. State of the art optical point-to-point links achieve <1Tb/s (NASA TBIRD experiment: 200Gb/s S2G, Starlink claim: 200Gb/s ISL).

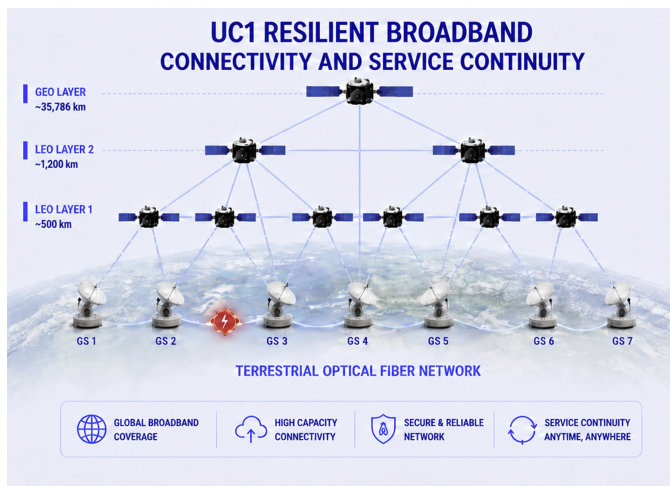


Fig a.2: Use case 1, UC1

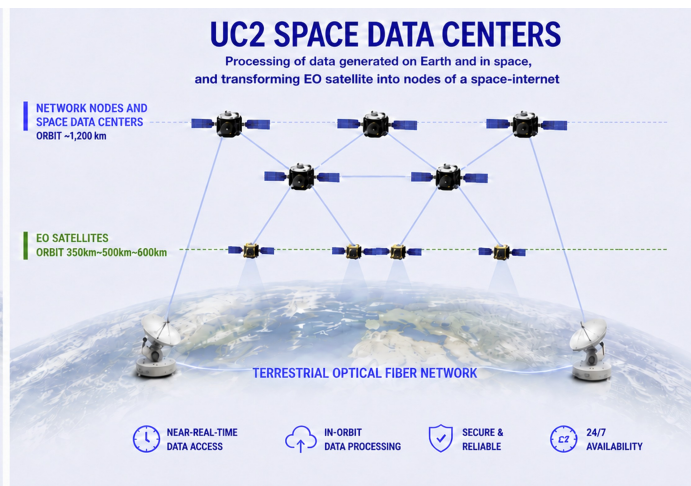


Fig a.3: Use case 2, UC2

OPTERA-Ob has thirteen task objectives Ox.y (associated to task Tx.y) and assessed through KPIs KPIx.y (KPI5.y are lab-validated, the others are simulation-validated).

O1.1 Project coordination: coordinate and monitor the scientific, administrative, financial, and ethical aspects of the project. KPI1.1: $\geq 90\%$ milestones on time; delays < 1 month; mitigation actions within 30 days.

O1.2 Dissemination and exploitation: promote communication, dissemination, and exploitation of results, maximizing scientific and social impact. KPI1.2: 1 website, 10 publications, 3 public reports, 1 conference workshop, 1 open-access repository, 1 follow-up project proposal submitted.

O2.1 Requirements and scenarios: define system assumptions, scenarios, use cases, and performance requirements. KPI2.1: quantitative specification of reference scenarios and system/network requirements.

O2.2 Architecture definition: define the reference space-ground system, identifying the main segments, their functions, and their interconnection interfaces. KPI2.2: architecture addressing 100% of O2.1 requirements.

O3.1 Channel models: develop modular models for G2S, S2G, ISLs, and fiber links, including attenuation, turbulence/scintillation, pointing, acquisition, and tracking (PAT) errors, phase noise, Doppler, Kerr nonlinearity, and noise. KPI3.1: parametrizable simulation-ready channel-model framework.

O3.2 Optical system design: design a coherent optical transmission architecture for satellite free-space optical (FSO) links with aperture diversity, adaptive power control and transparent multi-hop forwarding. KPI3.2: space-compatible coherent optical link, supporting transparent multi-hop transmission and probabilistic amplitude shaping (PAS) dual polarization (DP) 64-QAM at 120 Gbd per wavelength in the C band (≥ 3 hops without optical-electrical-optical (OEO) regeneration), with dynamic optical power control, up to 10 wavelengths.

O3.3 DSP design: develop the digital chain for adaptive coherent optical links, including PAS DP-QAM, synchronization, carrier/timing recovery, equalization, space-time block codes and digital beam combining, soft demapping, FEC, interleaving, automatic repeat request (ARQ), and erasure recovery. KPI3.3: ≥ 1 Tb/s per wavelength, fine rate granularity, post-FEC bit error rate (BER) $\leq 10^{-15}$.

O4.1 Routing algorithms: design algorithms that exploit the dynamics of low-Earth orbit (LEO) constellations to ensure low-latency and reliable path selection. KPI4.1: packet delivery ratio (PDR) $> 99\%$ under nominal traffic conditions ($> 5\%$ di service availability vs greedy), PDR $\geq 90\%$ under 10% link failure or overload conditions, routing convergence time ≤ 1 s, at least 30% reduction in route changes or link reconfigurations compared with purely delay-optimal routing, keeping delay degradation below 10%.

O4.2 Handover and switching algorithms: design optical handover/link switching under visibility, pointing, and intermittent-connectivity constraints. KPI4.2: handover failure rate $\leq 1\%$ in nominal visibility and pointing conditions; handover decision latency ≤ 50 ms.

O4.3 SDN design: develop a telemetry-driven SDN framework for multi-orbit space-ground networks with adaptive routing, transparent WDM forwarding, and AI-assisted control. KPI4.3: $> 40\%$ header overhead reduction vs Segment Routing over IPv6 (SRv6), e.g., from ~ 800 – $1,500$ B to < 500 – 900 B on long multi-hop paths; path enforcement time (PET) upon path computation < 5 ms; multi-agent coordination time < 500 ms.

O5.1 Experimental testbed: develop a lab-scale testbed for space-ground optical coherent communication systems and networks.

KPI5.1: commercial off-the-shelf (COTS) and proprietary subsystems compatibility; integration of optical subsystems, network nodes, and FSO links; real-time emulation of FSO impairments (e.g., scintillation, PAT error and Doppler/frequency offset); symbol rate up to 40 Gbd; modulation format up to 64QAM, WDM channels up to 10.

O5.2 Optical transmission experimental validation: experimentally validate a DP coherent optical communication system operating in single-wavelength and WDM configurations (up to 10 channels), integrating advanced DSP for dynamic rate adaptation under varying FSO channel attenuation conditions, with the goal of demonstrating high transmission capacity and system scalability.

KPI5.2: single channel bit rate 250 Gb/s; 10 WDM channels (2.5 Tb/s).

O5.3 Optical satellite network experimental demonstration: demonstrate end-to-end space-ground optical networking architecture through an integrated experimental setup, including dynamic routing, handover, and telemetry-driven adaptation in a realistic multi-node scenario, validating the designed protocols, routing strategies, and link management mechanisms under realistic operating conditions. KPI5.3: demonstration of the two UCs over ≥ 2 optically impaired links including both commercially available 100G/400G links and the coherent optical communication system by O5.2, validating dynamic routing, handover and telemetry-driven adaptation with routing recovery time ≤ 2 s, handover success rate $\geq 90\%$, service interruption time ≤ 1 s, and

≥20% improvement in service continuity or delivered data volume compared with static/non-adaptive operation.

State of the Art and beyond

Context and use cases. The new space economy is accelerating LEO constellations for future NTN [1]. Optical ISLs/feeder links increase throughput and reduce latency. Recent studies proposed EO-LEO integrations [2], while SDCs are emerging for in-orbit storage, EO processing, and AI workloads [3,4,5]. **Innovation:** provide innovative solutions for the two UCs: (UC1) resilient broadband/service continuity for underserved areas and disaster recovery; and (UC2) always connected SDC nodes for space- and terrestrial-generated data based on high-capacity optical links.

Optical channel models. Optical ISLs are mainly affected by pointing errors, satellite jitter, tracking inaccuracies, phase noise, and Doppler, while atmospheric impairments are generally absent [6]. G2S and S2G optical links also suffer atmospheric attenuation, turbulence/scintillation, cloud coverage, aerosols, beam wandering, and adaptive-optics imperfections [7]. **Innovation (T3.1):** develop integrated channel models for optical ISLs and feeder links, combining time-varying FSO, fiber (including high power optical amplifiers (HPOAs)), and PAT dynamics.

Optical satellite communication systems. Optical satellite links offer large bandwidth, narrow beams, and favorable size, weight, and power (SWaP) compared with RF systems [8]. Current systems mainly use direct detection and moderate data rates [9], despite increasing operational and experimental capacity [10,11]. Coherent detection and DSP are mature in terrestrial fiber, where 800 Gb/s-class wavelengths are already targeted [12], but satellite FSO adoption remains limited by turbulence, Doppler, fast power variations, pointing constraints, and SWaP [13]. **Innovation (T3.2):** develop coherent WDM optical transmission for satellite FSO, combining adaptive modulation/coding, multi-aperture diversity, DSP, optical power control, link-budget-aware design, and transparent multi-hop forwarding without OEO regeneration.

DSP chain for FSO satellite links. Coherent fiber systems rely on mature DSP and soft-FEC, but space-ground optical links introduce impairments, which affect synchronization, carrier recovery, rate adaptation, interleaving, and packet recovery [14,15]. **Innovation (T3.3):** define an adaptive DSP/FEC chain for coherent optical space-ground links, including QAM/PAS, synchronization, carrier recovery, space-time block codes, digital beam combining, soft demapping, FEC, interleaving, retransmissions, and erasure recovery.

LEO routing algorithms. LEO routing must handle continuously changing topologies. Geometry-aware approaches exploiting Walker-Delta regularity provide efficient low-latency routing solutions [16], while AI-based routing based on graph neural networks (GNNs) and deep reinforcement learning (DRL) is emerging for dynamic NTN scenarios. The number of optical inter-satellite terminals per satellite affects routing performance, reconfiguration rate, and scalability [17]. **Innovation (T4.1):** investigate time-aware and AI-assisted routing strategies for optical NTN constellations, targeting low latency, service continuity, reduced ISL reconfiguration, and realistic constraints on onboard complexity and optical-terminal availability.

Handover and switching for satellite network. Existing handover algorithms, often designed for access links, do not fully address optical G2S connectivity, site diversity, multiple concurrent connections, and pointing constraints [18]. Current resource-allocation approaches for data off-loading often use high-level channel models and rarely consider inter-shell optical off-loading [19]. **Innovation (T4.2):** develop optical handover, link-switching, and resource-allocation strategies jointly optimized with time-aware routing, visibility windows, site diversity, and time-varying link quality.

SDN design for satellite network. Terrestrial control-plane solutions, including NETCONF/YANG and segment routing, are not directly applicable to dynamic ground-satellite networks because of intermittent path availability, control-plane flooding, and header-overhead constraints [20]. Agentic AI and interoperability protocols are emerging in telco networks but are not yet established for ground-satellite optical systems [21]. **Innovation (T4.3):** develop a lightweight telemetry-driven SDN framework for transparent WDM optical forwarding, compact source routing, selective dissemination of link-state and impairment information, and AI-assisted coordination of remediation actions.

Experimental demonstration of optical satellite systems and networks. Coherent optical satellite/FSO demonstrations have progressed from tens-of-Gb/s to >100 Gb/s links using advanced modulation and DSP [22], and to WDM-enabled Tb/s-class aggregate capacities [23]. Rate-adaptive transceivers and automatic gain-control solutions have been demonstrated under dynamic conditions [24,25], but atmospheric turbulence and fast fading remain major limitations [26], requiring various mitigation techniques [27,28]. Experiments with COTS coherent transceivers show that current FEC and adaptation mechanisms remain critical under atmospheric dynamic fading [29]. **Innovation (WP5):** move beyond point-to-point coherent FSO demonstrations by experimentally integrating adaptive coherent WDM links with routing, handover, telemetry-driven control, and network-level service-continuity mechanisms in a laboratory space-ground optical network testbed.

OPTERA's methodology

OPTERA adopts an objective-oriented methodology. The global OPTERA-Ob is achieved through task objectives and five milestones (MSs). OPTERA is organized into five work packages (WPs), tasks Tx.y, task objectives Ox.y, and KPIs KPIx.y.

Five MSs define the achievement of important goals (see Section c). The PERT diagram (Fig a.4) highlights the relation among objectives, MSs, and OPTERA-Ob. O2.1 and O2.2 define the requirements and architecture and mark milestone MS1, providing a clear framework for system-oriented O3.y and network-oriented O4.y objectives. The achievement of system-oriented objectives O3.y marks MS2, unlocking the experimental activities for O5.1, O5.2 which, in turn, mark MS3. The achievement of system-oriented objectives O3.y and MS2 provide essential information to network-oriented objectives O4.y, whose achievement marks MS4. Together, system and network objectives, MS3 and MS4 unlock the final lab-demonstration O5.3. Together with the achievement of O1.1 and O1.2, O5.3 marks the achievement of the final milestone MS5 and of OPTERA-Ob.

The consortium brings together four complementary competences needed to bridge mature fiber-based optical technologies and space-oriented network architectures: optical fiber systems and networks, and RF satellite systems and networks. Fig a.5 shows that OPTERA has all the necessary expertise to develop a high-capacity optical satellite communication network. The methodology is multidisciplinary, bringing together different and complementary expertise, from physical-layer technologies to network- and control-layer functions. In detail, system-oriented tasks require expertise in fiber propagation and impairments (O3.y, O5.y – CNR,

SSSA), FSO propagation and impairments (O3.y – POLITO, CNR), optical transmission systems and subsystems (O3.2, O5.y – CNR, SSSA), signal design, DSP and equalization (O3.2, O3.3 – CNR, SSSA, UNIVPM), and error correction coding (O3.2, O3.3 – CNR, UNIVPM, POLITO). Network-oriented tasks require expertise in LEO constellation design (O4.1, O4.2, O5.3 – POLITO, CNR), network-level design (O4.y, O5.3 – SSSA, POLITO, CNR), routing algorithms (O4.1 – POLITO, CNR, SSSA), handover protocols (O4.2 – CNR, POLITO), and SDN (O4.3 – SSSA). OPTERA combines these complementary competences with project management, dissemination, and exploitation experience. A more detailed description of the skills of each unit is in Section c. The soundness of OPTERA's methodology relies on a staged validation strategy that progressively connects physical-layer innovation with network-layer operation and lab-validation. In particular, OPTERA investigates how the physical-layer constraints of Tb/s coherent optical transmission can be translated into timely network-layer decisions, enabling adaptation to rapidly evolving satellite topologies, with the resulting cross-layer approach validated through laboratory experiments. This rationale is reflected in the MSs structure: MS2 and MS3 address the physical-layer system, MS4 validates its integration with the network layer, and MS5 demonstrates the resulting architecture in a laboratory environment.

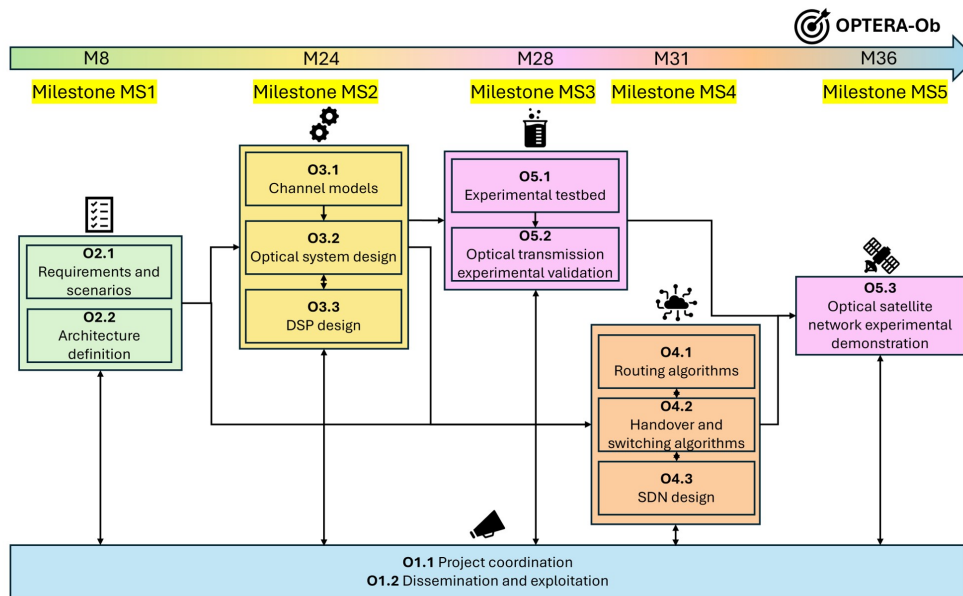


Fig a.4: PERT diagram

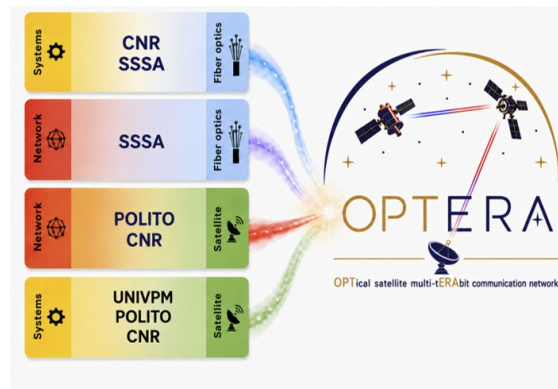


Fig a.5: Expertise and complementarity of the OPTERA units

Risk and mitigation strategies legend: Rn,K,J is Risk number n with likelihood K and severity J, with K and J in {L=Low, M=Medium, H=High}

R1,M,M-WP1-Delays in milestones and deliverables. Mitigation: regular WP and plenary meetings will support early detection of delays; resources will be reallocated and critical tasks prioritized to prevent cascading effects.

R2,L,H-WPall-Inaccurate financial estimates may lead to budget deviations. Mitigation: semiannual budget reviews will identify deviations early; effort and expenditures will be adjusted within the call rules to preserve project objectives.

R3,L,H-WP1-The PI or associated PIs transfer to a different institution or become unable to continue. Mitigation: the measures allowed under Call D.D. 2298 will be applied. Each task involves multiple senior researchers, avoiding single points of failure in scientific leadership.

R4,H,H-WPall-Difficulties in recruiting personnel with the required expertise. Mitigation: increase the effort of A.1 personnel for critical activities (maximum 20% allowed); use consultancy/external expertise. Improve contract attractiveness by offering higher salaries (fewer positions); reduce effort by downscaling lower-priorities activities (e.g., study of transparent fiber-FSO links, multi-aperture spatial diversity).

R5,M,H-WP5-Devices and equipment unavailability due to procurement delays or conflicts with other experimental activities. Mitigation: multiple suppliers, early identification of alternative components and backup instruments; early-scheduling of orders and measurements (compressed campaigns); supply-chain risk monitoring; external equipment loans.

R6,M,M-WP4-Dynamic routing and handover mechanisms may introduce service interruption or control-loop instability under rapidly varying channel conditions. Mitigation: adaptive thresholds, hysteresis mechanisms, telemetry filtering, and policy-based routing constraints will be used; validation will progressively increase impairment dynamics.

R7,H,H-WP5-High optical powers (<43 dBm) in the laboratory may damage commercial optical components, including fibers, connectors, and optical devices, which are not tailored to support high power. Mitigation: connector cleaning/pigtail splicing, fiber-bending control, insertion of optical splitters and attenuators. If high-power operation is impractical, high-power effects will be emulated through highly nonlinear fibers.

R8,M,H-WP3,5- The nonlinear effects in HPOA are not accurately modeled due to lack of available data, possibly affecting the mitigation techniques. Mitigation: T3.1 effort update to develop and validate new nonlinear models at the expenses of multi-aperture spatial diversity; for WDM operation, amplify each channel independently and mux before transmission.

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b. Credibility of the pathways to achieve the results and expected impacts specified in the work programme, as well as the likely scale and relevance of the project's contributions; Adequacy and quality of the measures aimed at maximizing the results and expected impacts on society, on the territory, and on the advancement of knowledge

The credibility of OPTERA's impact relies on a clear and realistic link between the project's activities, the expected results, concrete dissemination and exploitation measures, and the expected outcomes targeted by the work programme.

Technological Maturity

In terms of technological maturity, OPTERA starts from an initial TRL of 2, since the enabling technologies are individually established but not yet integrated into a programmable coherent optical space-ground network. The final laboratory demonstration, including coherent WDM transmission with rate adaptation, emulated FSO impairments, routing, handover, and telemetry-driven adaptation, is expected to bring the integrated concept to TRL 4. The lab-validation will also test the use of COTS devices, making the proposed solution more accessible, scalable, and closer to practical exploitation.

Alignment with the impacts specified in the work program

OPTERA is fully aligned with the guiding principles of the PRIN 2026 programme and Italian "Piano Triennale della Ricerca 2026-2028". Indeed, the project addresses a fundamental research challenge of national relevance: the development of scientific and technological foundations for high-capacity optical satellite communication systems and networks. Due to its complexity and multidisciplinary nature, this challenge cannot be effectively addressed by a single institution but requires the coordinated collaboration of research units with complementary expertise in optical fiber systems and networks, satellite systems and networks, signal processing, coding, control, routing, and network management. In this sense, OPTERA strengthens the national research system by fostering collaboration among universities and research institutions, promoting the integration of scientific competences distributed across the territory, and creating a coherent consortium able to address both system-oriented and network-oriented

objectives. The project is therefore consistent with the territorial and collaborative dimension promoted by PRIN 2026, as well as with its guiding principles of scientific excellence, impact, feasibility, and efficiency in project implementation.

Additionally, consistently with the bottom-up nature of the PRIN programme and its spirit of “free scientific research”, OPTERA is conceived as a researcher-driven project, originating from an autonomous scientific vision shared by the proposing units. The project aims to advance the fundamental understanding and experimental validation of coherent optical space-ground networking, beyond the constraints of a predefined industrial roadmap or a mission-oriented technological procurement need.

Alignment with national and international strategic priorities

OPTERA is well aligned with major Italian, European and global strategic priorities in digital connectivity, space communications, resilience and technological sovereignty. In particular, OPTERA directly addresses needs highlighted in several policy and strategic documents; selected examples are shown in Fig b.1.

At global level, OPTERA contributes primarily to sustainable development goal (SDG) 9, Industry, Innovation and Infrastructure, by supporting advanced and resilient communication infrastructures. In addition, OPTERA may also contribute to SDGs 3, 4, 8, 10 and 16, by enabling more reliable connectivity for healthcare, education, economic development, inclusion, and secure public services. At European level, the European Commission’s “White Paper on Europe’s digital infrastructure needs” emphasizes the importance of high-quality and reliable connectivity across the Union, including rural and remote areas, as well as the relevance of terrestrial and non-terrestrial network convergence to ensure uninterrupted connectivity. OPTERA is also aligned with the objectives of IRIS², the EU Secure Connectivity Programme, which targets resilient satellite communications, including broadband connectivity for emergency response and for areas with limited terrestrial network coverage. Furthermore, Horizon Europe supports research and innovation in satellite telecommunications, with 68 M€ dedicated funding in 2026 and 2027. OPTERA is fully aligned with the European Research Council (ERC) PE7 panel, “Systems and Communication Engineering”, with specific relevance to the PE7_6 Communication technology and high-frequency technology, PE7_7 Signal processing, and PE7_8 Networks subsectors.

At national level, OPTERA is consistent with the priorities of the “Piano Strategico Space Economy”, which includes several programmes in telecommunications, and with ASI’s “Piano Triennale delle Attività 2024-2026”, which aims at the development of optical satellite networks with the HydRON project.

Credibility of the project implementation, expected results, and exploitation potential, adequacy and quality of the measures for maximizing results and impacts

OPTERA’s project activities are structured in a coherent and well-organized work plan, supported by a consortium with highly complementary expertise across the key scientific, technological, and application domains addressed by the project. Clearly defined KPIs enable continuous monitoring of technical progress and project achievements. Realistic risk mitigation measures have been identified to address potential implementation challenges. The credibility of the expected results is further strengthened by concrete dissemination and exploitation measures, including public technical reports and software releases, peer-reviewed publications, participation in international conferences, and the preparation of follow-up project opportunities. Overall, these elements make OPTERA’s expected results credible, realistic and likely to be achieved, disseminated and further exploited beyond the project duration.

The credibility of the pathways to achieve the results and expected impacts specified in the work programme is further supported by letters of support from leading Italian and international industrial stakeholders: two complementary letters from Thales Alenia Space Italia reflecting distinct industrial perspectives (product-oriented and strategic research-and-technology) within the same organization (Figs b.2-b.3), Aerospazio Tecnologie srl (Fig b.4), TIM S.P.A (Fig b.5), MPB Communications Inc. (Fig b.6), and OSMIUM Italy srl (Fig b.7). These letters highlight both the relevance and urgency of the use cases addressed by OPTERA, as well as the strength and credibility of the consortium in delivering the proposed outcomes and their likely.

OPTERA is expected to generate exploitable knowledge, methods, and tools for companies and start-ups active in optical communications, satellite networks, space-ground connectivity, and network automation. The availability of validated concepts, open software tools, and COTS-based laboratory demonstrations may facilitate technology transfer, industrial prototyping, and follow-up collaborations, lowering the entry barrier for future commercial development and exploitation. Stakeholder engagement will be an integral component of OPTERA’s strategy. The letters of support provided by international companies provide an initial indication of external interest, to be further consolidated through continued interactions for industrial uptake. OPTERA results will be disseminated to relevant standardization bodies and industry forums (ESTOL, ETSI, Broadband Forum, ITU, OIF, IETF, ONF), which will facilitate the future exploitation of these results in a broader perspective for the benefit of the entire European industry and research centres.

Adequacy and quality of the measures maximizing results and impacts

OPTERA’s impact is grounded in concrete project outcomes: a lab-validated coherent optical space-ground transmission framework, impairment-aware control and routing methods, and a reusable experimental platform for future NTN research. These results directly support resilient connectivity in emergency scenarios and underserved areas such as islands, mountain regions, and remote communities, while strengthening national expertise in strategic technologies for optical NTNs, including coherent WDM, DSP/FEC, dynamic handover, and SDN-based orchestration. The project will also provide reusable modeling and validation tools for future ASI, ESA, Horizon Europe projects and industrial collaborations, and will train young researchers with interdisciplinary skills across optical communications, space networking and programmable network control. This impact is credible because OPTERA focuses on reproducible lab demonstration, rather than on short-term space deployment.

Additionally, OPTERA partners are involved in other research projects and activities that offer strong complementarities and synergies (Fig b.8).

Measures to maximize impact

OPTERA will maximize its results and expected impacts through an integrated strategy combining scientific dissemination, open and reproducible research, exploitation of results, stakeholder engagement, communication towards non-specialist audiences, and training of young researchers, outreach via social media platforms and project website.

These measures are designed to ensure that the project outcomes do not remain confined to individual research tasks, but are transferred to the scientific community, made visible to relevant industrial and institutional stakeholders, and positioned as a basis for future research and innovation activities in the field.

The effectiveness of these measures will be monitored throughout the project using specific indicators, including scientific publications, conference contributions, public reports, software releases, dissemination and outreach activities, training activities, and follow-up initiatives. This monitoring will allow the consortium to adapt its dissemination, exploitation, and communication strategy during the project, ensuring that OPTERA's results and impacts are maximized in a realistic and measurable way.

Impact on society and territory, and advancement of knowledge

OPTERA addresses clear societal challenges, including the continuous growth of global data traffic and AI-driven computing, the need for resilient broadband connectivity in emergency situations and for disaster recovery, the increasing demand for EO strategies, and the need to ensure continuous connectivity and critical digital services also in remote areas of the country (as addressed by national and international strategic priorities in Fig b.1). The expected societal impact of OPTERA will be maximized by explicitly linking the project's scientific and technological results to UCs of direct societal relevance. Indeed, these UCs will guide the definition of requirements, performance indicators, and validation scenarios, ensuring that the project results remain connected to concrete societal needs.

OPTERA has relevant impacts on the national territory. On the one hand, OPTERA will boost the advancement of knowledge throughout the territory thanks to the distributed and collaborative structure of the consortium, which integrates complementary competences from different institutions, regions, and research fields. Indeed, OPTERA brings together the biggest research institute of Italy with nationwide presence (CNR), a leading university for advanced studies and applied research located in Pisa (Scuola Superiore Sant'Anna), and two polytechnic universities located in Turin and Ancona, as shown in Fig c.7 of Section c. On the other hand, OPTERA aims at providing relevant solutions to grant continuous connectivity and EO strategies for Italy, where the heterogeneous geography of the country (including mountainous areas, islands, rural regions, and remote communities) combined with frequent exposure to natural hazards such as wildfires, floods, landslides, and earthquakes, creates a strong need for resilient and rapidly deployable communication infrastructures.

The advancement of knowledge is maximized through targeted scientific dissemination and exploitation actions, with a project task with 17 person-months (PMs) dedicated to these activities (T1.2 "Dissemination and exploitation"). OPTERA promotes high-quality publications in peer-reviewed international journals and conferences, as well as invited talks, tutorials, and the organization of a dedicated workshop within an international conference, aimed at fostering cross-fertilization between different research communities (optical systems, optical networks, coding, RF satellite systems). OPTERA also promotes open and reproducible research practices ("as open as possible, as closed as necessary") through the release of software tools and public technical reports, and the publication of open-access scientific papers. At the same time, OPTERA considers the protection of exploitable results through intellectual property rights, including patent applications, and enabling future start-up initiatives.

OPTERA also maximizes the advancement of knowledge through training and human-capital development. The project provides young researchers, PhD students, postdoctoral researchers, and technical staff with a multidisciplinary research environment at the intersection of optical communications, photonic systems, networking, coding, and satellite communications. Through close interaction with senior researchers and professors, participation in joint research activities, internal seminars, cross-unit technical meetings, and research visits to partner institutions in Italy (as shown in Fig c.7 of Section c), OPTERA supports the training and formation of researchers and engineers in a strategic technological domain. Furthermore, OPTERA will recruit 2 PhD students and create 9 additional one-year positions for early-career researchers, thereby directly investing in scientific excellence and the advancement of knowledge. Additionally, OPTERA will organize a workshop on topics addressed by the project, which will boost the dissemination of the results and the advancement of knowledge.



WHITE PAPER

How to master Europe's digital infrastructure needs?

Beyond terrestrial connectivity, further investments are required for the integration of advanced satellite services providing complementary solutions for backhaul, device connectivity in remote areas not covered by terrestrial technologies or to ensure service continuity in case of crisis or disaster relief.

Finally, recent geopolitical developments highlighted the importance of security and resilience of infrastructures against both human-made and natural hazards, as well as the complementary role of terrestrial, satellite and submarine connectivity solutions, for uninterrupted availability of service under all circumstances. In a rapidly changing security landscape, a strategic Union-

IRIS² at a glance

The IRIS² Satellite Constellation is the European Union's third flagship, addressing long-term challenges of EU's security, safety and resilience by offering enhanced connectivity services to governmental users.

The new multi-orbital constellation of 290 satellites will combine the benefits offered by Medium Earth Orbit (MEO) and Low Earth (LEO) satellites. It is set to provide secure connectivity services to the EU and its Member States as well as broadband connectivity for governmental authorities, private companies and European citizens, while ensuring high-speed internet broadband to cope with connectivity dead zones.

#EUSpace

Horizon Europe
Work Programme 2026-2027

7. Digital, Industry and Space

- HORIZON-CL4-2026-SPACE-03-31: Digital enablers and building-blocks for Earth Observation and Satellite telecommunication for Space solutions (Space Partnership) – RIA – 12M€ – 2026
- HORIZON-CL4-2026-SPACE-03-32: Preparing demonstration missions for Earth Observation and Satellite telecommunication for Space solutions (Space Partnership) – IA – 26M€ – 2026
- HORIZON-CL4-2027-SPACE-03-33: Digital enablers and building blocks for collaborative Earth Observation and Satellite telecommunications for Space solutions (Space Partnership) – RIA – 4M€ – 2027
- HORIZON-CL4-2027-SPACE-03-34: Preparing demonstration missions for collaborative Earth Observation and Satellite telecommunication for Space solutions (Space Partnership) – IA – 26M€ – 2027

PIANO STRATEGICO SPACE ECONOMY - Quadro di posizionamento nazionale

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Attività ASI in ambito ESA – piano Triennale attività 2024-2026 ASI

- HydRON (Sviluppo di future reti satellitari ottiche ad alta capacità):

HydRON (High throughput Optical Network), progetto incluso linea di programma strategico ARTES sulle comunicazioni ottiche e quantistiche (ScyLight) mira a realizzare il dimostratore della prima rete di trasporto completamente ottica al mondo con capacità di terabit/sec, basata su satelliti multi-orbita, in modo da estendere nello spazio senza soluzione di continuità le reti terrestri in fibra ottica.

Il progetto consentirà lo sviluppo e la validazione delle richieste tecnologie chiave da parte dell'industria europea. Supporterà la prossima generazione di missioni di telecomunicazioni spaziali istituzionali e commerciali, che richiedono capacità di comunicazione avanzate.

Fig b.1: Global, European, and Italian strategic priorities



ThalesAlenia Space Italia S.p.A.
Via Saccomano, 24
00131 Roma
Italia
Tel.: + 39 06 4151 1
Fax: + 39 06 4190 575

Rome, May, 19th 2026

To whom it may concern

Thales Alenia Space is glad to support the OPTERA project. The challenges it tackles (higher-capacity feeder links, inter-satellite connectivity, and tighter integration between space and ground segments) are ones we see coming up repeatedly as we work on next-generation satellite systems, and optical communication is increasingly central to how we think about addressing them.

What stands out about OPTERA is that it does not treat these as purely physical-layer problems. The project takes coherent optical transmission and adaptive signal processing seriously, but it also engages with the architectural questions: how do you make these links reliable, how do you integrate them into programmable network infrastructures, and what does this actually look like in realistic use cases? That combination of depth and breadth is exactly what makes research useful to us as system designers.

The topics OPTERA addresses (optical feeder links, inter-satellite links, adaptive transmission chains, resilient terrestrial and non-terrestrial network architectures) are not abstract. They map directly onto design decisions we face today and will face more acutely as demand for secure, high-capacity, and flexible satellite communications continues to grow across institutional, commercial, and strategic applications.

Yours faithfully,

Lorenzo Simone

Head of Processing, Navigation & Power Products

Thales Alenia Space – Italia

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Capitale Sociale € 204.067.999,00 int. vers. Codice Fiscale ed Iscrizione al Registro Imprese di Roma n° 02101600480 - P. IVA n° 00991340909
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Fig b.2: Thales Alenia Space support letter



Roma, 19/05/2026

All' attenzione dei proponenti del progetto,

Thales Alenia Space è lieta di supportare il progetto OPTERA proposto per un bando PRIN (Progetto di Ricerca di Interesse Nazionale) avente come tema lo sviluppo di tecnologie ottiche per la prossima generazione di comunicazioni satellitari. Le sfide che affronta (collegamenti feeder a capacità più elevata, connettività inter-satellite e un'integrazione più stretta tra segmenti spaziali e terrestri) sono quelle che vediamo emergere frequentemente nel nostro lavoro sui sistemi satellitari di prossima generazione, e la comunicazione ottica assume un ruolo sempre più centrale nell'economia spaziale.

Leonardo Mazzini,

Chief Technical Officer,
Thales Alenia Space Italia, Rome

[OPEN]

Thales Alenia Space Italia S.p.A. (Società con unico socio, soggetta alla attività di direzione e coordinamento congiunta di Thales S.A. e di Leonardo S.p.A.)
Sede Legale: Via Saccomano, 24 00131 Roma - Tel.: + 39 06 4151 1 - PEC: thales@spaceitalia.it
Capitale Sociale € 204.067.999,00 int. vers. Codice Fiscale ed Iscrizione al Registro Imprese di Roma n° 02101600480 - P. IVA n° 00991340909
Modello: 8823043-000-146-11884

Fig b.3: Thales Alenia Space support letter



RAPOLANO TERME (SI), 25/05/2026

ATT TO: PRIN 2026 Evaluation Committee

SUBJECT: LETTER OF SUPPORT FOR THE OPTERA PROJECT

Dear Dr. Stella Civelli,

I would like to convey Aerospazio Tecnologie interest in and support for the project *Optical satellite multi-terabit network (OPTERA)*. The initiative addresses several key challenges associated with the evolution of next-generation satellite communication systems, particularly in the areas of high-capacity optical links, coherent optical transmission technologies, adaptive communication strategies, and transparent multi-hop networking architectures. These research directions are highly relevant in view of the increasing demand for scalable and efficient space-based communication infrastructures.

The project objectives are regarded as both ambitious and technologically significant, especially with respect to the development of innovative coherent optical transmission solutions capable of supporting very high data rates and adaptive operation under dynamic channel conditions. The proposed investigation of transparent optical networking approaches without onboard OEO regeneration is also considered particularly valuable for reducing payload complexity and improving overall network efficiency.

Aerospazio Tecnologie operates in the field of laser communication terminals for space applications. Its core activities include a novel testbed where the laser terminals may be tested for environmental compatibility and interoperability. Over the years, the company has developed expertise in providing test and qualification services, with applications related to satellite technologies, aerospace systems, optical communications, and advanced telecommunication infrastructures. The themes addressed within OPTERA are therefore closely connected with the company's technological interests and long-term strategic objectives.

Over many years, Aerospazio Tecnologie has followed the scientific and technological activities of Scuola Superiore San'Anna (SSSA) and interacted with its researchers through several collaborations. These interactions have highlighted the institution's strong research capabilities, technological expertise, and effective approach to translating advanced research into innovation. We also recognize the availability at SSSA of a state-of-the-art optical communications laboratory with advanced capabilities in coherent transmission and optical networking. On this basis, we believe that the OPTERA team has the expertise, infrastructure, and scientific background needed to successfully address the proposed challenges.

For these reasons, Aerospazio Tecnologie considers OPTERA a highly valuable research initiative and expresses its support for the proposed project.

Yours sincerely,

Fabrizio Scortecchi
Administrator
Aerospazio Tecnologie srl

Sede legale: Via Provinciale Nord 42/a, 53040 Rapolano Terme, Siena - Capitale Sociale Euro 100.000 int. - Registro Imprese SIENA, C.F., P.IVA 01008700528 - R.E.A. 113538

Fig b.4: Aerospazio Tecnologie srl support letter



Prof. Roberto Garelli
Dipartimento di Elettronica e Telecomunicazioni (DET)
Politecnico di Torino
Corso Duca Degli Abruzzi 24, Torino
+39 0110904106
roberto.garelli@polito.it

Torino, May 25th 2026

Subject: Letter of Intent - OPTERA

On behalf of TIM SpA, I am pleased to express our interest in supporting the PRIN 2026 OPTERA project. OPTERA addresses a strategic topic for the future of communications: the development of fully optical satellite networks and their integration with next-generation terrestrial and non-terrestrial infrastructures. From the perspective of a TIM, the major Italian telecommunications operator, the challenges addressed by OPTERA are highly relevant, as future networks will increasingly require high-capacity, flexible, and resilient architectures capable of ensuring seamless integration across heterogeneous communication domains.

As Head of Technology Innovation, Standard & IPR at TIM, where I am involved in the development and evolution of advanced communication networks and services and in 3GPP standardization activities related to NTN- TN integration and interoperability. I also personally know some members of the consortium and have had very positive collaborations with them in the past. I am confident that this project can generate valuable scientific and technological outcomes, also thanks to the strong collaboration among the partners and the effective dissemination of its results.

For these reasons, I believe OPTERA addresses research directions of clear strategic interest for the future evolution of communication networks, and I am glad to express my support for this initiative.

Best regards

Head of Technology Innovation, Standard & IPR
Gabriele ELIA
Gabriele.elia@telecomitalia.it

TIM S.p.A.

Sede legale: Via Gaetano Negri, 1 - 20123 Milano
Sede secondaria e Direzione Generale: Corso d'Italia, 41 - 00198 Roma
Codice Fiscale: 04488410010
Iscrizione al Registro A.E.E. IT08022000000799
Capitale Sociale € 11.677.002.855,10 interamente versato

Fig b.5: TIM S.p.A. support letter



To Dr. Stella Civelli
CNR, ISTITUTO DI ELETTRONICA E DI INGEGNERIA DELL'INFORMAZIONE E DELLE
TELECOMUNICAZIONI
Via Girolamo Caruso, 16, 56122 – Pisa – Italy

Subject: Support letter, Civelli Application to PRIN

May 27, 2026

Dear Dr. Civelli,

On behalf of MPB Inc., I am pleased to express our support for the project entitled: **"Optical satellite multi-TERabit communication network (OPTERA)"**.

MPB Communications Inc. is pleased to express its support for the OPTERA project, which addresses highly relevant challenges in the field of high-capacity optical communications for future space-ground and satellite network infrastructures.

MPBC has long-standing expertise in the development of advanced optical technologies, including optical amplification solutions for demanding communication and photonic applications. In this context, we recognize the increasing strategic importance of optical satellite communications, where high-capacity, reliable and resilient links are expected to play a key role in the evolution of next-generation space and non-terrestrial networks.

OPTERA proposes a timely and technically relevant approach by combining coherent optical transmission, WDM technologies, adaptive signal processing, transparent optical forwarding and programmable network control for dynamic space-ground communication scenarios. The project's focus on realistic optical impairments, high-capacity links, and resilient end-to-end connectivity is particularly aligned with the need to further study and validate enabling technologies for future satellite communication systems.

From MPBC's perspective, OPTERA offers a valuable opportunity to investigate the role of optical amplification technologies in advanced satellite communication architectures. In particular, the project can contribute to a better understanding of the requirements, constraints and performance trade-offs associated with the use of optical amplifiers in free-space optical links, ground-to-satellite communication scenarios, inter-satellite connectivity and transparent optical networking concepts.

We consider the OPTERA consortium to be well positioned to deliver the proposed objectives, thanks to its complementary expertise in optical transmission systems, photonic technologies, satellite communications, networking and experimental validation. The planned laboratory demonstration and system-level analysis will provide relevant insights for the maturation of technologies that may support future high-capacity and resilient optical satellite communication infrastructures.

MPBC therefore welcomes the objectives of OPTERA and supports its ambition to advance programmable coherent optical networks for space-ground communications. We believe that the project can generate results of strong scientific and technological relevance, while contributing to the development of the European and international ecosystem for secure, resilient and high-capacity satellite connectivity.

1/2

MPB Communications Inc.

MPB Communications Inc. (MPBC) is a privately owned Canadian company headquartered in Montreal, with electronic test centres in Ottawa and Edmonton. Established in 1977, MPBC is an OEM supplier of fiber-optic laser sources and amplifiers serving the space and ground telecommunications, industrial, biomedical, and research markets. Over the years, MPBC has continuously innovated and reinvented its products to address increasingly demanding technical requirements and enable scientific and technological breakthroughs. Notably, MPBC's visible fiber-laser technology was used by 2014 Nobel laureates Dr. Stefan Hell and Dr. Eric Betzig in the development of super-resolved fluorescence microscopy.

Space Photonics is one of MPBC's three main product portfolios and continues to grow year after year. MPBC's space heritage has been developed since its early years through participation in demonstrators and space payloads supported by Canadian Space Agency and European Space Agency programs, including ESA's PROBA-2 launched in 2009 and still operating to this day. Building on these successful space missions, MPBC has, over the past decade, developed and produced all-fiber lasers and optical amplifiers for military, governmental, and commercial applications. In the optical satellite communications market, MPBC supplies flight-proven, TRL-9 optical amplifier units for satellite constellations and programs such as the Space Development Agency, ESA's HyDRON, and Telesat Lightspeed. To date, MPBC has delivered more than 600 systems to leading laser communication terminal manufacturers, with production and deliveries ramping-up. In parallel, MPBC is actively extending the range, applicability, output power and wavelength coverage of its space optical amplifier technology, including the development of 100 W high-power amplifiers and 1,000 W optical ground-station amplifier systems for optical feeder-link applications targeting multi-terabit-per-second satellite data transmission.

MPBC is also a leader in the development of laser guide star systems. Its first system was installed in May 2015 and brought online at the European Southern Observatory's Very Large Telescope in Chile. MPBC is currently in development of higher power laser guide star solutions, partnering with Italian companies LEOS Solutions, Srl and MicroGate Srl, with an innovative system that utilizes a non-resonant second harmonic generator. Through this collaboration, MPBC is responsible for the fiber-based Raman amplifier module.

Over the years, MPBC has received numerous awards, including the Canadian Awards for Business Excellence, the Innovation and Entrepreneurship Award, the Canada Export Award, and the Photonics Circle of Excellence Award, recognizing the company's leadership in advanced photonics technologies.

Yours sincerely,

Dr. Emile Haddad
Program manager
Space and Photonics
MPB Communications Inc.

2/2

Fig b.6: MPB Communications Inc. support letter

Osmium Italy s.r.l.
Corso Castellidardo 30/A
10129 Torino (Italy)
Partita IVA: 12914450015



Letter of Support – OPTERA Project

I am Juan José Grosso, founder and CEO of Osmium Italy SRL (OSMIUM), a cybersecurity company with the mission of safeguarding the aerospace, defence and automotive industries' digital transformation and enabling the secure use of digitalization, connectivity, and SW-based systems to guarantee the safety, integrity, and resilience of its customer's missions and operations.

I am pleased to express my support for the OPTERA project, which addresses a strategic topic for the future evolution of satellite communications, namely the development of fully optical satellite networks.

From the perspective of space infrastructure security and resilience, the challenges addressed by OPTERA are highly relevant, as future satellite systems will increasingly require communication architectures that are not only high-capacity and flexible, but also robust, resilient, and secure, where optical technologies can offer important advantages.

I personally know some members of the consortium and I am familiar with their expertise in satellite systems technologies: I am confident that this project can generate valuable scientific and technological outcomes and contribute to research directions of clear strategic interest for the space sector.

For these reasons, I am glad to express OSMIUM's support for this initiative.

Juan José Grosso, CEO
Osmium Italy SRL



Osmium Italy s.r.l. - Corso Castellidardo 30/A 10129, Torino (Italy), Partita IVA: 12914450015

Fig b.7: OSMIUM Italy srl support letter

Common Partners: CNR
ANTERRA – MSCA-DN: Inter-satellite communication and synchronization
Link to OPTERA: The project involves channel modeling and performance assessment of the optical inter-satellite link, as well as synchronization algorithms, in the context of a LEO constellation.
Common partners: CNR
ALLEGRO- HORIZON-CL4-2022; GA 101092766; 01/2023 - ongoing: ALLEGRO aims at designing and validating a novel E2E sliceable, reliable, and secure architecture for next-generation optical networks, achieving high transmission/switching capacity.
Link to OPTERA: Terrestrial network supporting high-capacity and adaptation through optical coherent WDM system developed in ALLEGRO may be integrated within OPTERA.
Common Partners: POLITO, UNIVPM
MASSCON – ESA: design of ground station to satellite communication systems
Link to OPTERA: The project addresses aspects of satellite network design and operation, providing technical background on NTN architectures and related communication challenges.
Common Partners: POLITO
HANDING OVER – ESA: routing, handover, and radio resource management algorithms for VLEO constellations
Link to OPTERA: The project addresses mechanisms to support connectivity management in dynamic NTN scenarios characterized by satellite mobility, time-varying visibility, and changing link conditions.
Common Partners: POLITO
STARGATE – ESA: study and evaluation of waveforms for 6G NTN
Link to OPTERA: The project addresses aspects of space communication systems, contributing to the development of next-generation satellite systems

Fig b.8: Other projects by OPTERA partners with synergy opportunities

c. Organization of the project with respect to the proposed objectives, the timeframes considered necessary for the completion of the project, and the required resources; appropriateness of the composition of the research team, also in terms of complementarity and the time commitment indicated in the proposal, in relation to the proposed objectives; capacity and role of each operational unit, also within the research team, in implementing the work plan; appropriateness of the planning in the management of project activities: consistency between costs and activities, adequate temporal allocation of resources, clarity and justification of budget items, efficiency, management transparency, and financial sustainability

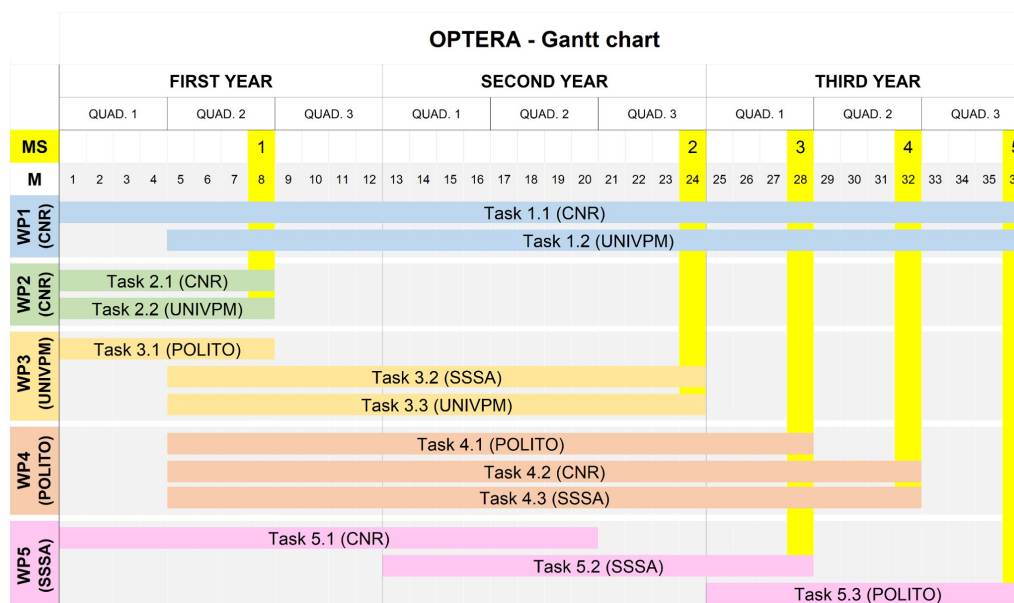


Fig c.1: GANTT chart

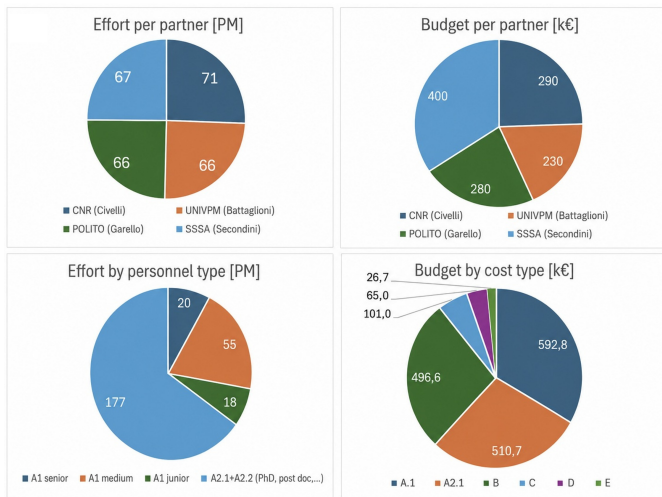


Fig c.2: Effort and budget distribution by unit or cost type

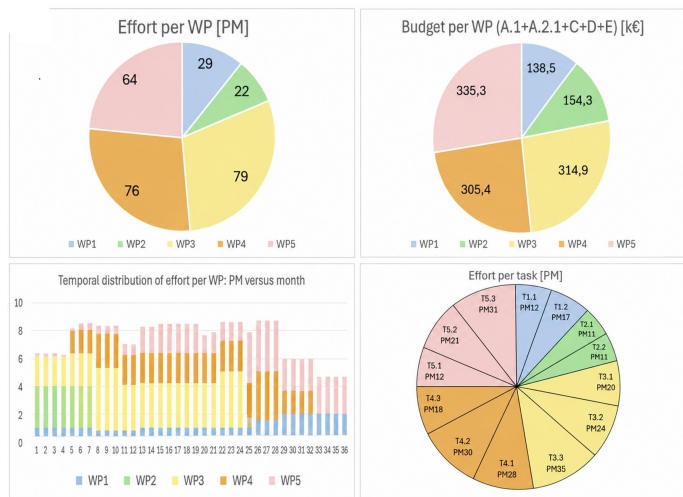


Fig c.3: Effort and budget distribution by WP or time

Organization of the project

The project is organized into five WPs and thirteen tasks (task Tx.y belongs to WPx), each dedicated to Ox.y (Section a). Five MSs mark the end of a set of activities, unlock the continuing of the project and assess the progress of the project. Four deliverables are also planned. The timeline of each activity (expressed in months M) is in the GANTT in Fig c.1, reporting also the WP/task leader. The relation among activities and objectives is in the PERT in Section a. The WPs, tasks, MSs, and deliverables are described below. Figs c.2-c.3 show the effort in PMs per partner, the effort per WP, and the effort per task, highlighting that the effort per WP is similar for WPs 3,4,5, and that the effort in PM is well distributed among partners.

WP1-Project Management (Leader: CNR)

Task 1.1 Project management

T1.1 ensures effective overall management of the project across partners, guaranteeing compliance with applicable regulations and to the work plan, proper coordination of activities, and timely achievement of MSs and deliverables. CNR leads the task, ensuring effective organization and management of all project aspects, in collaboration with UNIVPM, SSSA and POLITO. Addresses MS5.

Task 1.2 Dissemination and exploitation

T1.2 disseminates project results through peer-reviewed publications, conference presentations, a dedicated conference workshop, a dedicated project website, and open-access repositories, in line with open-science principles. A Python software suite, released via GitHub, will collect selected simulation tools. T1.2 identifies exploitation pathways, including research and industrial follow-ups, technology-transfer opportunities, and contributions to standardization activities. UNIVPM leads; UNIVPM and POLITO coordinate dissemination and scientific outreach; SSSA supports exploitation planning; CNR supports project visibility. Responsible for deliverables D1, D2, D3, D4. Addresses MS5.

WP2-System and Network Architecture and Requirements (Leader: CNR)

Task 2.1 Scenarios and system requirements

T2.1 defines the representative use cases, operational scenarios, main system assumptions, and functional/performance requirements of OPTERA. The task identifies the relevant network segments, node roles, traffic flows, constellation and ground-station assumptions, and service-level targets. Two UCs guide the definition: UC1 broadband and service-continuity scenarios, including remote areas, disaster recovery, and TN/NTN convergence; and UC2 SDCs for AI-computing and EO satellite processing and continuous connectivity. CNR leads the task; UNIVPM identifies coding and reliability requirements; POLITO contributes system-level, networking, and routing requirements; SSSA defines optical-link and testbed-related requirements. Addresses MS1.

Task 2.2 Reference architecture definition

T2.2 defines the reference architecture addressing the scenarios and requirements identified in T2.1. The task specifies the organization of the coherent optical space-ground network, including intra-satellite, G2S and S2G links, and multi-hop optical connectivity, the main interfaces among ground, space, optical-transmission, networking, control, and testbed segments, and the architectural assumptions needed by WP3, WP4, and WP5. UNIVPM leads the architecture definition; CNR contributes optical terminal and system constraints; POLITO contributes networking, routing, handover, and control-plane aspects; SSSA contributes coherent optical implementation and experimental testbed constraints. Addresses MS1.

WP3 – Optical Coherent System Model and Design

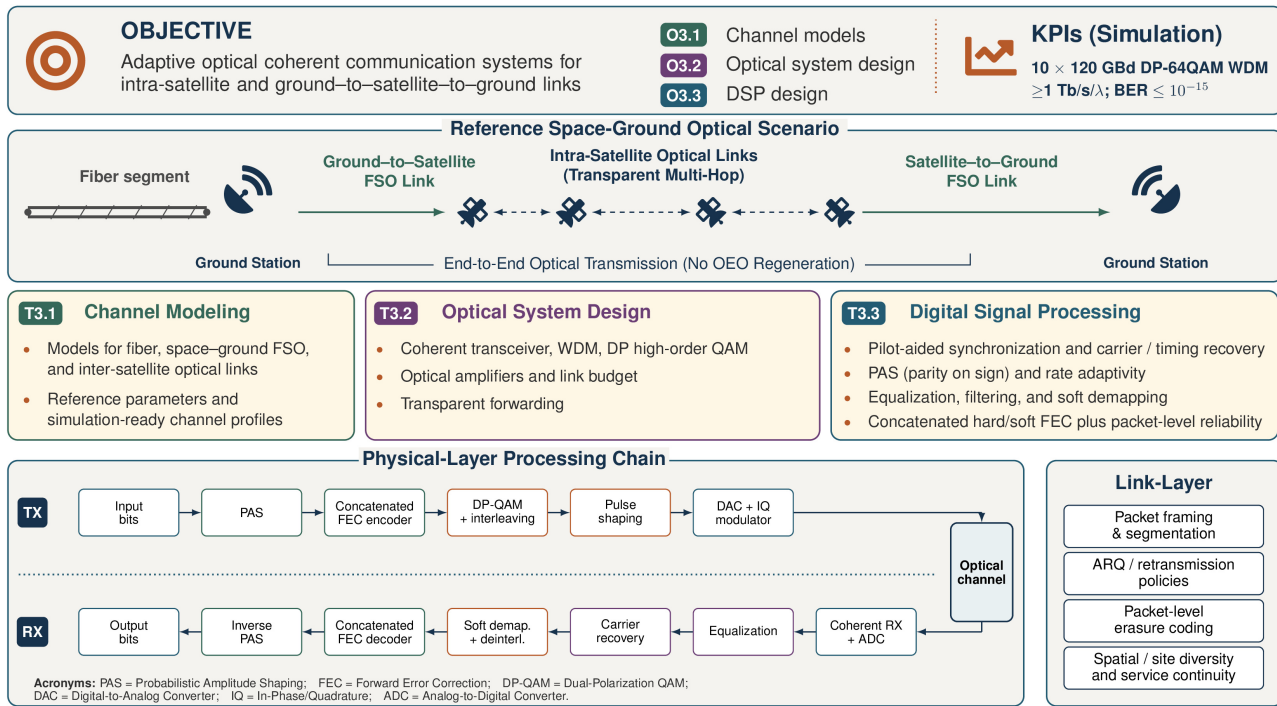


Fig c.4: description of WP3

WP3-System Model and Design (Fig c.4) (Leader: UNIVPM)

Task 3.1 Channel model

T3.1 develops a consistent modeling framework for the main propagation impairments affecting optical space-ground, ISL and fiber segments, including atmospheric effects, PAT errors, phase noise, Doppler, turbulence, and fiber nonlinearities. POLITO leads the task and space-ground FSO modelling, SSSA addresses fiber channels and amplifiers, CNR models ISL optical channels, and UNIVPM translates the channel models into simulation inputs for DSP, coding, and link-level performance assessment. The units will coordinate closely to ensure a common structure and harmonized parameters across the different channel segments. Addresses MS2.

Task 3.2 Optical system design

T3.2 covers physical-layer design of end-to-end point-to-point satellite optical links, targeting high-data-rate coherent transmission with high spectral efficiency and adaptation to time-varying channels via flexible modulation/coding/shaping. Includes space-qualified optical hardware dimensioning (lasers, amplifiers, modulators, photodetectors, coherent receivers), dynamic optical power control for SNR adaptation, multi-carrier (WDM) for aggregate capacity increase, and multi-aperture spatial diversity to mitigate scintillation, beam wandering, and speckle. Develops link power budgets and transparent multi-hop optical links enabling end-to-end transmission across satellites without OEO regeneration, reducing latency and onboard complexity. SSSA leads the task and develops the optical system with CNR; UNIVPM covers integration with T3.3; POLITO covers space-qualified constraints and dynamic optical power control. Addresses MS2.

Task 3.3 Digital signal processing

T3.3 develops the digital chain for coherent optical links, including DP-QAM/PAS modulation, pilot-aided synchronization, carrier and timing recovery, equalization, space-time block codes and coherent digital beam combining for aperture diversity, soft-demapped FEC, symbol interleaving, ARQ policies, and packet-level erasure recovery. UNIVPM leads the task and coding/reliability; POLITO evaluates ARQ, erasure and diversity trade-offs; CNR covers terminal/link-budget assumptions; SSSA contributes coherent-transceiver and implementation constraints. Addresses MS2.

WP4 – Programmable Networking for Space–Ground WDM Optical Systems

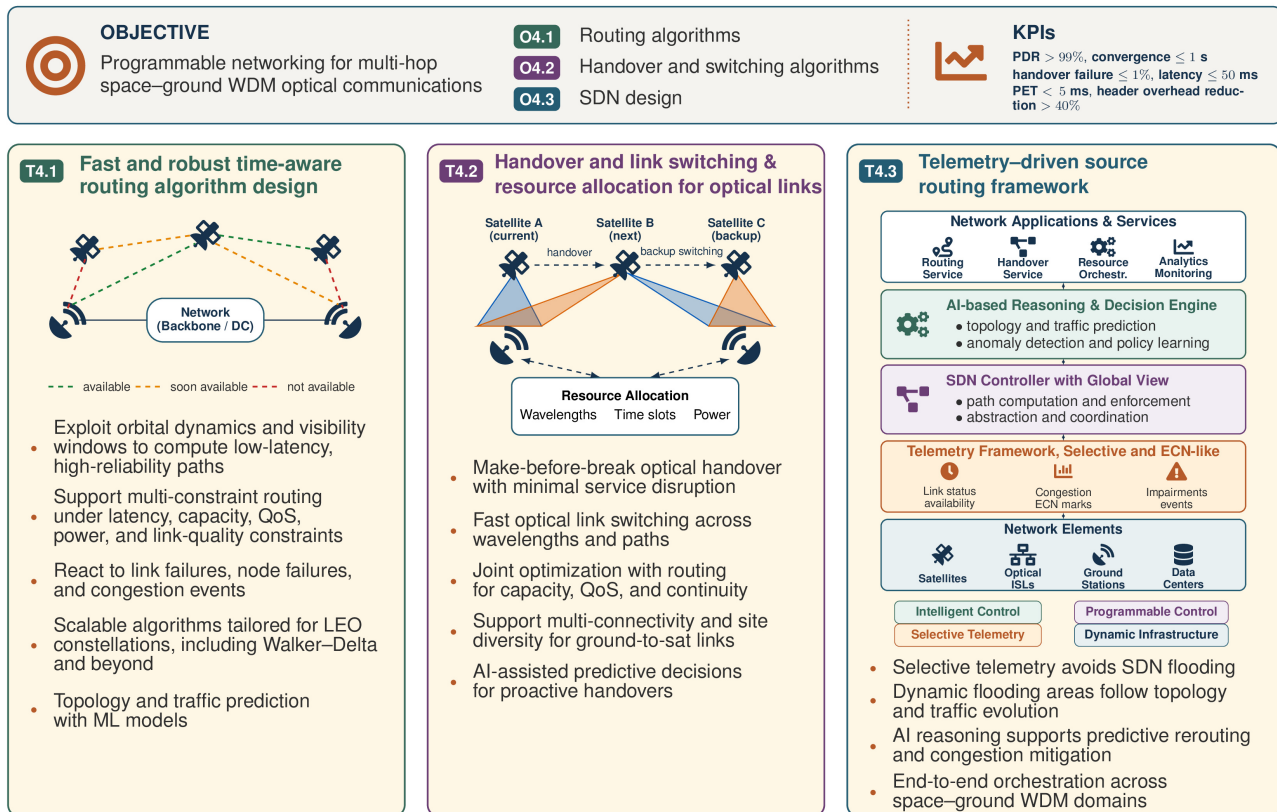


Fig c.5: description of WP4

WP4-Programmable Networking for Space–Ground Systems (Fig c.5) (Leader: POLITO)

Task 4.1 Fast and robust time-aware routing algorithm design

T4.1 aims to develop time-aware routing algorithms that explicitly account for the predictable dynamics of LEO satellite constellations, exploiting predictable orbital dynamics, link availability windows, and physical-layer models from WP3, to compute efficient and stable routing decisions, ensuring low latency, high reliability, and robustness against failures and traffic overload conditions, under rapidly changing topologies. The effect of different numbers of onboard terminals is assessed in terms of performance-robustness trade-offs. POLITO leads the task and investigates different routing approaches, including innovative solutions as DiscoRoute, considering Walker–Delta constellations. CNR contributes to scalable routing design and performance evaluation, UNIVPM and SSSA supports implementation analysis and validation. Addresses MS4.

Task 4.2 Handover and link switching & resource allocation for optical links

T4.2 focuses on the design of classical and AI-based methods and protocols for space–ground node association, possibly with heterogeneous satellite orbits, considering the dynamic network topology, the time-varying G2S link capacities and the availability of satellite-to-satellite multi-hop links for offloading of information received from feeder links. T4.2 aims to define a distributed multiple-input and multiple-output (MIMO) network between satellites and ground stations, enabling multiple connections per node to increase throughput, provide path and site diversity, and improve resilience against adverse atmospheric conditions. CNR leads the task and develops the main algorithms, with POLITO supporting their integration with T4.1 routing outputs and SSSA providing practical constraints and network-level validation. Addresses MS4.

Task 4.3 Telemetry-driven source routing framework

T4.3 develops a telemetry-driven source routing framework for heterogeneous ground-satellite networks, enabling transparent WDM forwarding and multi-hop space–ground paths without flooding-based protocols. T4.3 exploits ultra-dynamic and adaptive link-segment lists, compressed path-encoding mechanisms, and predictive rerouting using link/weather/congestion telemetry. Agent-to-agent (A2A) and model-context protocol (MCP) support distributed agents at ground nodes for diagnosis, optimization and control. SSSA leads the task and develops the framework in collaboration with POLITO and CNR. Addresses MS4.

WP5 – OPTERA EXPERIMENTAL DEMONSTRATION

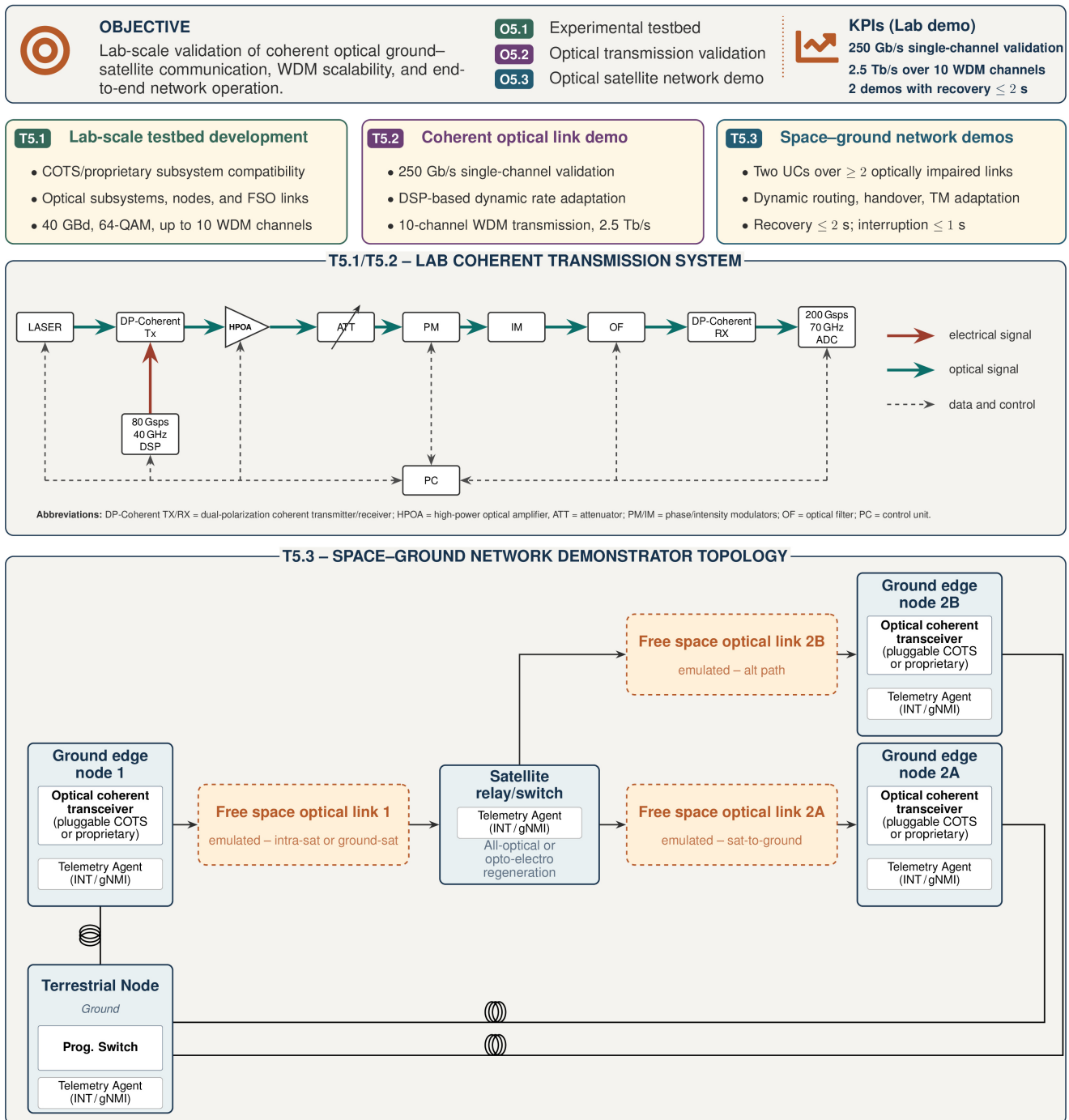


Fig c.6: description of WP5

WP5- OPTERA experimental demonstrations (Fig c.6) (Leader: SSA)

Task 5.1 Development of a lab-scale testbed

T5.1 develops a highly automated laboratory measurement setup for OPTERA's experimental validation in T5.2 and T5.3, based on COTS and proprietary devices. For T5.2, the system shall include a digital-to-analog converter (DAC), a laser in the C-band, a DP-IQ modulator, a HPOA, a FSO channel emulator (based on T3.1 and made of a variable attenuator, a phase modulator, and an amplitude modulator), a programmable optical filter based on liquid crystal on silicon (LCoS) technology, a DP coherent receiver, an optical local oscillator, and a four-channel analog-to-digital converter (ADC). The system shall incorporate DSP at both ends before DAC and after ADC. The system should also support WDM operations. For T5.3, the same system is extended to incorporate multiple network nodes. T5.1 includes the identification, selection, and procurement of the necessary equipment. CNR leads the task and covers FSO propagation emulation. SSSA and UNIVPM supports the development and integration, contributing to channel impairments emulation. Addresses MS3.

Task 5.2 Lab demonstration of coherent optical satellite link

T5.2 validates high data rate optical coherent communication in a laboratory environment at SSSA. The target data rate is 250 Gb/s using a PAS 64-QAM signal. Large bandwidth equipment such as a 70 GHz bandwidth-200 Gs/s oscilloscope and a 40 GHz bandwidth-100 Gs/s DAC enable reliable and repetitive measurements. Channel variations are emulated using appropriate statistics for fast control in term of phase and amplitude variation. Nonlinear effects introduced by HPOA are minimized by appropriate signal

design and WDM allocation. T5.2 integrates the DSP features developed in WP3 and demonstrates dynamic rate adaptation. WDM operation is implemented with up to 10 channels in the C-band, providing an overall capacity of up to 2.5 Tb/s. SSSA leads the task and integrates subsystems and performs experimental validation in collaboration with CNR, POLITO, and UNIVPM whose role will be mainly in the DSP refining and optimization. Addresses MS3.

Task 5.3 Experimental demonstration of space-ground networks

T5.3 is the final task of the OPTERA and is devoted to the experimental validation of the two UCs. The optical system is as in T5.2. The demonstrator for UC1 includes: programmable switching and routing nodes based on both commercial 100 G/400 G Ethernet and proprietary platforms; coherent optical transmission subsystems; telemetry and control modules; and optically impaired links emulating realistic space communication conditions. Performance is evaluated through appropriate metrics for communication reliability, service continuity, and network resilience in critical and highly challenging operational scenarios, e.g., pre-FEC BER or generalized mutual information (GMI), link margin, end-to-end throughput, packet loss/misordering. The demonstrator for UC2 emulates always-connected SDC nodes exchanging high-rate data streams and distributed AI/LLM workloads over the optical network infrastructure. Performance is evaluated as communication reliability, service continuity, end-to-end latency, delivered data volume, workload interruption time, distributed processing completion time, and AI workload availability. POLITO leads the task, SSSA leads the experimental integration of network functions, CNR and UNIVPM contribute to system integration and performance evaluation. Addresses MS5.

Milestone MS1 System and network architecture and requirements defined – MS1 marks the end of WP2 and the definition of the network architecture and specification required for the continuation of the project. Feeds WP3 and WP4. Addressed by T2.1 and T2.2.

Milestone MS2 System model and design concluded – MS2 marks the end of WP3 and the design of end-to-end, point-to-point, adaptive optical coherent systems for high-capacity satellite communications. Feeds WP4 and WP5. Addressed by T3.1, T3.2 and T3.3.

Milestone MS3 Point-to-point optical link experimentally validated – MS3 marks the end of T5.2 and the successful lab-based emulation of satellite point-to-point coherent communications. Feeds T5.3. Addressed by T5.1 and T5.2.

Milestone MS4 Programmable networking framework for space-ground systems defined – MS4 marks the end of WP4. Feeds WP5. Addressed by T4.1, T4.2 and T4.3.

Milestone MS5 Network experimental demonstration completed and end of the project – MS5 marks the end of WP5 and the end of project. MS5 is the successful lab-based demonstration of a programmable network for space-ground communications based on high-capacity optical coherent systems. Addressed by T1.1, T1.2, and T5.3.

Deliverable D1 – First year report – public report – M12 – D1 collects all activities carried out in the first year of the project, highlights progress against planned objectives and KPIs and identifies any deviation from the original work plan with mitigation measures.

Deliverable D2 – Second year report – public report – M24 – D2 collects all activities carried out in the second year of the project, highlights progress against planned objectives and KPIs and identifies any deviation from the original work plan with mitigation measures.

Deliverable D3 – Final report – public report – M36 – D3 is the final report of the project. D3 provides a comprehensive overview of all activities carried out throughout the entire duration of the project. D3 highlights progress against the planned objectives and KPI, and assesses the overall achievements and outcomes. D3 also outlines future perspectives, including potential follow-up projects and exploitation of project results.

Deliverable D4 – Simulation platform – public software – M36 – D4 is a Python software suite, released as open-source via GitHub, which integrates a selection of the software tools developed and used throughout the project. D4 provides a user-friendly environment for running simulations, reproducing key results, and supporting further research and development.

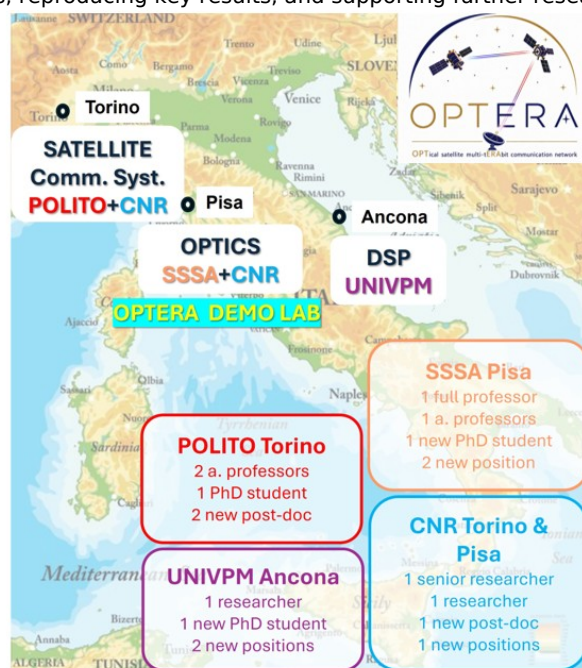


Fig c.7: Geographic distribution of the OPTERA units

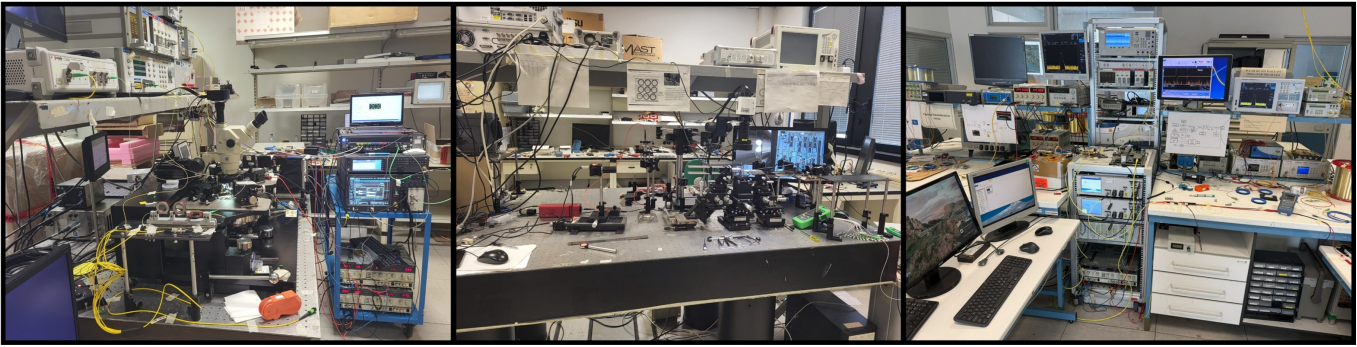


Fig c.8: SSSA laboratory (joint lab with CNIT PNTLab)

Composition and complementarity of the research team in relation to the proposed objectives

The research team comprises four units: one research center and three universities, distributed across three Italian cities: Torino, Pisa and Ancona. Each location contributes a distinctive area of expertise (Fig c.7), creating a geographically distributed and scientifically complementary consortium that strengthens OPTERA's multidisciplinary approach, essential for OPTERA's objectives, which require the joint integration of optical systems and networks, satellite communication systems, advanced DSP techniques, and programmable networking.

The Pisa unit, involving SSSA and CNR, provides strong capabilities in optical fiber communications, from physical-layer modeling and system design to network-level aspects and experimental validation, supported by a long-standing collaboration demonstrated by joint publications. Also, SSSA hosts a well-established laboratory for photonic technologies and optical networks, jointly operated with CNIT PNTLab for more than 20 years, and also hosting some CNR equipment <https://pntlab.cnit.it/> (Fig c.8). This facility allows to perform OPTERA's experimental validation with a relatively small budget (discussed below). The Turin unit, with POLITO and CNR, contributes expertise in satellite communication systems at both physical and network layers. UNIVPM adds consolidated competences in DSP, coding theory and related techniques. Overall, this structure ensures a geographically distributed but closely coordinated team, with the expertise required to effectively implement OPTERA. The expertise of each unit within the team and in relation to the objectives is shown in Fig c.9. A more detailed description follows in Fig c.10.

The research team is heterogeneous, comprising both senior and junior personnel, including professors, researchers, early-career researchers, and PhD students (see Figs c.2-c.7).

Gender balance is not yet satisfactory (25% women) but OPTERA is led by a female PI, confirming the consortium's commitment to inclusive leadership. OPTERA will seek to improve gender balance during implementation.

Appropriateness of the composition of the team in terms of time commitment

Figs c.2-c.3 highlights that (i) the effort per task/WP is well distributed, allocating more effort to core-WPs: WP3, WP4, and WP5, (ii) the distribution of effort among the unit is well distributed, (iii) most of effort is on early-career researchers (e.g., PhD students, post-docs) under the guidance of tenured personnel (A.1).

Capacity and role of each unit in implementing the workplan

The capacity and role of each operational unit in implementing the work plan are detailed in Fig c.10, highlighting, for each unit, the WP and task leadership, the main areas of expertise by topic, the WPs and tasks in which this expertise is used, the MSs to which it contributes, and selected key references supporting the declared expertise.

Appropriateness of the planning and management transparency

Figs c.2 and c.3 present OPTERA's effort and budget distribution from different perspectives: i) the effort and budget per task/WP is well distributed, with core-WP being the largest, (ii) the effort is coherently distributed in time confirming a coherent and sustainable allocation of resources (also considering that A2.1 personnel are expected to join after the initial phase and finish before the project closure), (iii) the effort and budget per partner is well distributed, given that SSSA covers costs for the lab-validation (CNIT consultancy, equipment and consumables), (iv) most of effort is on younger researchers (e.g., PhD students, post-docs) under the guidance of A.1 personnel, (v) most of the budget is for personnel, but a large part (A.1 and A2.2) is not in the requested funding (co-funding for A.1 and zero-cost for the project for A.2.2, in accordance with the PRIN 2026 call), (vi) budget is primarily allocated to personnel, while laboratory instrumentation accounts for a limited share of the budget, thanks to the existing experimental facilities available in Pisa, (vii) general expenses (item B), corresponding to € 497k, are sufficient to cover the project's overhead needs, including the institutional share retained by the participating institutions.

Details of budget items (excluding employed personnel and overhead) are in Fig c.11.

OPTERA's results are owned by the project partner that performs the work from which such results arise. If a result is created jointly by two or more partners and it is not possible to identify or separate their respective contributions, that result is jointly owned by the partners that contributed to it. The same principle for inventions, patents applications and granted patents.

OPTERA will adopt an "as open as possible, as closed as necessary" strategy for all project results. Open-science practices are maximized, for example through open-access peer-reviewed publications, public technical deliverables and reports, and the release of software tools.

Research data will be managed according to FAIR principles of Findability, Accessibility, Interoperability, and Reusability, ensuring secure storage, accessibility, interoperable formats, and adequate documentation to support reuse and reproducibility.

Competency legend: Skilled ✓, Master 🦾

	CNR Civelli	UNIVPM Battaglioni	POLITO Garello	SSSA Secondini	Target objectives	Key expertise
Free-space optics propagation models	✓		✓	✓	O3.1, O5.1	PAT; Doppler; scintillation
Optical fiber systems and subsystems	🦾			🦾	O2.1, O2.2, O3.1, O3.2, O3.3,	Fiber propagation; amplifiers; coherent TRX
Error correction codes	✓	🦾	✓		O2.1, O3.2, O3.3,	LDPC; soft decoding;
Optical networks				🦾	O2.2, O4.1, O4.2, O4.3,	Optical nodes; telemetry;
Satellite communication systems and networks	🦾		🦾		O2.1, O2.2, O4.1, O4.2, O4.3, O5.3	Handover; LEO routing; EO satellites; constellations
Optical communication lab-validation	✓			🦾	O5.1, O5.2, O5.3	HPOA; proprietary TRX; programmable attenuators
Digital signal processing	🦾	🦾	✓	🦾	O2.1, O3.3, O5.2	PAS; CPR; nonlinearity mitigation; equalization
Project management and dissemination	🦾	🦾	🦾	🦾	O1.1, O1.2	Project coordination; workshop organization; peer-reviewed publications

Fig c.9: Expertise of each unit within the team and in relation to the objectives

Unit	Lead	Expertise	WP/Task	MS	Key Reference
CNR	WP1, WP2, T1.1, T2.1, T4.2, T5.1	Project management, coordination, and dissemination	WP1	MS5	[allegro] , [persuader] , [rigoletto], [anterra]
		Optical fiber systems, digital communication, probabilistic shaping, fiber nonlinearity compensation, fiber propagation, carrier phase recovery	WP2, WP3, T5.1, T5.2	MS2, MS3	[Civ2023ILT], [Civ2023OFC], [Civ2024JLT], [Civ2025ILT], [Civ2026arxiv]
		Satellite communication networks, handover algorithms, error-correcting codes, inter-satellite link, DSP, interference management, resource allocation, routing algorithms	WP2, T3.1, T3.3, T4.1, T4.2	MS4	[Tar2026WONS], [Tar2025SOFTCOM], [Tar2025MAS], [Tar2024PIMRC], [Tar2024TQE]
UNIVPM	WP3, T1.2, T2.2, T3.3	Project management, coordination and dissemination	WP1	MS5	[econet], [masscon]
		Error-correcting codes, FEC design, hard/soft decoding, digital communications, satellite RF communications, PHY/MAC reliability mechanisms, ARQ/FEC interaction	WP2, T3.1, T3.3	MS2	[Bat2024ITC], [Bat2025ITAES], [Bat2025IA], [Bat2021JUSCN]
SSSA	WP5, T3.2, T4.3, T5.2	Project management, coordination, and dissemination	WP1	MS5	[quasar], [nqsti], [rigoletto], [cotone]
		Channel modelling, optical communication theory, optical system design, optical communication lab, carrier recovery, DSP	WP2, WP3, WP5	MS2, MS3, MS5	[Sec2015] [Poti2012], [Poti2015], [Sambo2013], [Sec2022]
		Optical networking, segment routing, SDN, networking for AI, agentic AI	WP4, T5.3	MS4, MS5	[Morro2025], [Cast2025], [Cast20252]
POLITO	WP4, T3.1, T4.1, T5.3	Project management, coordination, and dissemination	WP1	MS5	[masscon] [handing over]
		LEO constellations for internet broadband, space IoT, EO satellites, space data centers	WP2	MS1	[frisat], [Gar2022], [Gar2024], [Gar2026]
		Error-correcting codes, HARQ for NTN	T3.3	MS3	[Gar2023], [Gar2025acc]
		Satellite networks, LEO Routing, radio resource management, handover	WP4	MS3	[handing over] [Gar2025soft] [Gar2025ASHS]

Publications

[Civ2023JLT] Civelli et al., "On the nonlinear shaping gain with probabilistic shaping and carrier phase recovery", JLT (2023)

[Civ2023OFC] Civelli et al., "Probabilistic shaping methods for linear and nonlinear channels" OFC 2023, invited paper.

[Civ2024JLT] Civelli et al., "Sequence-Selection-Based Constellation Shaping for Nonlinear Channels" JLT 2024.

[Civ2025JLT] Civelli et al., "A new twist on low-complexity digital backpropagation" JLT 2025

[Civ2026arxiv] Civelli et al., "Nonlinearity Compensation for Coherent Optical Satellite Communications" Arxiv preprint 2603.08422

[Tar2026WONS] Hina et al., "QoS-aware Resource Management with Network Slicing and Beam Management", WONS Conference 2026.

[Tar2025SOFTCOM] Rivello et al., "Location-Based Radio Resource Management with MU-MIMO for VLEO-NTN Systems", SOFTCOM Conference 2025.

[Tar2025MAS] Tarable et al., "UAV/HAPS feeder link in W and D bands: a feasibility study", MAS 2025.

[Tar2024PIMRC] Ben Salem et al., "Uplink soft handover for LEO constellations: how strong the inter-satellite link should be", PIMRC 2024.

[Tar2024TQE] Tarable et al., "Rateless protograph LDPC codes for quantum key distribution", IEEE TQE, 2024.

[Bat2024ITC] Battaglioni et al., "Rate-compatible LDPC Codes based on Primitive Polynomials and Golomb Rulers," ITC, 2024.

[Bat2025ITAES] Battaglioni et al., "Design and Analysis of the Tail Sequence for Short LDPC-Coded Space Communications," TAES, 2025.

[Bat2025IA] Giuliani et al., "Telecommand Rejection Probability in CCSDS-Compliant LDPC-Coded Space Transmissions With Tail Sequence," IEEE Access, 2025

[Bat2021JUSCN] Battaglioni et al., "Effect of randomizers on the power spectrum excess of space telemetry signals," JUSCN, 2021.

[Sec2015] Secondini et al., "Optical Time-Frequency Packing: Principles, Design, Implementation, and Experimental Demonstration," JLT, 2015

[Poti2012] Poti et al., "Casting 1 Tb/s DP-QPSK Communication into 200 GHz Bandwidth," in ECOC 2012.

[Poti2015] Poti et al., "Sub-Nyquist field trial using time frequency packed DP-QPSK super-channel within fixed ITU-T grid," OE 2015

[Sambo2013] Sambo et al., "Software defined code-rate-adaptive Terabit/s based on time-frequency packing," OFC 2013

[Sec2022] Secondini et al., "New lower bounds on the capacity of optical fiber channels via optimized shaping and detection", JLT 2022

[Morro2025] Morro, et al "Disaggregated and transparent multi-band optical continuum across access, horseshoe aggregation, and metro IPoWDM networks", JOCN, 2025

[Cast2025] Castoldi, "Shaping the future of optical networks by integrating SDN, telemetry, and AI", JOCN, 2025

[Cast20252] Castoldi, "Programmable packet-optical network security and monitoring using DPUs with embedded GPUs", JOCN, 2025

[Gar2022] Garelio et al., "Sparse satellite constellation design for global and regional direct-to-satellite IoT services", TAES, 2022

[Gar2023] Garelio et al., "Performance improvement of space missions using convolutional codes by CRC-aided list viterbi algorithms", IEEE Access 2023

[Gar2024] Garelio et al., "On the use of mega constellation services in space: Integrating LEO platforms into 6G non-terrestrial networks", JSAC, 2024

[Gar2025acc] Garelio et al., "5G new radio for non-terrestrial networks: Analysis and comparison of HARQ and RLC ARQ performance over satellite links", IEEE Access 2025

[Gar2025soft] Benso et al., "Distributed On-Demand Routing for VLEO Constellations with 3-Terminal Inter-Satellite Links", IEEE SoftCOM 2025

[Gar2025ASMS] Garelio et al., "Advanced Routing Strategies for LEO and VLEO Constellations: Ensuring Polar Coverage", ASMS 2025

[Garelio2026] Garelio et al. "Deep Tech to Space: Space Data Centers and AI Revolution at the Edge", submitted to IEEE CM, arxiv, 2026

Projects

[allegro] HORIZON-RIA ALLEGRO project – Civelli was PI for CNR-IEIT unit, third party of CNIT – July 2024-July 2025

[persuader] PERSUADER – Italy-Israel R&D Cooperation Program MAE – April 2024-April 2026, Civelli was PI for CNR-IEIT unit

[rigoletto] PNRR-RESTART – Civelli was the point of contact for CNR units in RIGOLETTO project, Secondini was leader of Task 3.5 – January 2023-March 2026

[anterra] Anterra MSCA Project, Grant Agreement No. 101072363 – Tarable was responsible for a PhD on the subject "Intersatellite communication and synchronization

[econet] PNRR-RESTART – Battaglioni was the project coordinator of "Efficient and Reliable Communication Networks" (ECONET) project, Jul 2024 – Sep 2025. CUP J33C22002880001. Budget: 497.209,25€

[QUASAR] Italian Ministry of Defense PNRR Project, "QUAntum Safe netwoRk (QUASAR)", ongoing. M. Secondini as local PI of SSSA unit.

[nqsti] PNRR MUR Project "National Quantum Science and Technology Institute (NQSTI)", codice PE000000023 – CUPJ53C22003190001, 2023 – 2026. M. Secondini as leader of Spoke 7 for SSSA unit.

[cotone] Italian Ministry for Education University and Research (MIUR) FIRB Project – Futuro in Ricerca "COherent Terabit Optical NEtworks (COTONE)", 2012-2015. M. Secondini as PI of the project.

[masscon] ESA project "Multiple access sequences for satellite constellations without centralised management" (MASSCON), ARTES ITT AO10694, Jan 2023-present. Garelio CNIT Unit Leader, Battaglioni WP leader.

[handing over] European Space Agency project "Handover, Data Routing And Radio Resource Management For Very Low Earth Orbit (Vleo) Broadband Constellations (HANDING OVER)" ARTES 4.0 AT 3C.043, 2024-ongoing, Garelio POLITO Unit Leader

[frisat] European Space Agency project, "5G-NTN VLEO constellation for broadband D2D connectivity (FRIISAT)", 2025

[stargate] European Space Agency project "End-to-end demonstrator for 6G candidate waveforms (STARGATE)", ARTES 3F.041, 2026-ongoing, Garelio POLITO Unit Leader

Fig c.10: Capacity and role of each unit in implementing the workplan

	Unit	Justification	Budget
A2.1	CNR	- Two-year research contract "contratto di ricerca" (€ 96k), 24PM, focused on WP4 - One-year research contract "incarico di ricerca" (€ 30k), 12 PM, focused on WP3	€ 126k
A2.1	UNIVPM	- PhD student, 24 PM for OPTERA (€ 75k), for modeling and design of coherent optical space-ground links in WP2 and WP5 - Two one-year "incarichi di ricerca", 12PM € 30k each, focusing on the development, testing, and integration of WP3 simulation modules	€ 110k
A2.1	POLITO	- Two-year research contract "contratto di ricerca" (€ 100k), 24 PM, focused on WP3 and WP4 - One-year post-doctoral research contract "incarico postdoc" (€ 50k), 12PM, focusing on WP4	€ 150k
A2.1	SSSA	- PhD student, 24 PM for OPTERA, (€ 68k) focusing on WP3 and integration in WP5 - Two one-year research contract "incarico di ricerca", 12PM and € 28k each, focusing on WP4 and integration in WP5	€ 124k
C	SSSA	Equipment: 1x Keysight M8198A-002 Arbitrary Waveform Generator, 128 GSa/s, AXIe Module. The budget (list price: 258785.79) considers a depreciation period of 2.5 years during the project and an estimated 50% utilization rate https://www.keysight.com/used/us/en/signal-function/arbitrary-waveform-generators/m8198a-002-p08-101499 . This equipment will be used as programmable DAC and data generator for coherent transmitter	€ 64696
C	SSSA	1xHPOA (+43dBm output power-Amionics: AMN-AEDFA-43-R-FA), needed at the ground station as booster Quotation by Laser2000 dated May 15 th , 2026	€ 30267
C	SSSA	High-frequency RF amplifiers used as drivers for coherent optoelectronic IQ modulators	€ 6k
D	UNIVPM	Consultancy services to support WP3 implementation, including software modules, code optimization, simulation automation, data handling, and reproducibility of simulation tools. Under UNIVPM supervision, the consultancy will help avoid implementation bottlenecks in WP3 simulations and in the final open-access software platform	€15k
D	SSSA	CNIT consultancy service for using two CNIT devices in the jointly operated SSSA-CNIT lab for the WP5 experimental activities (Fig c.8). The cost is estimated for 80Gbps/40GHz dual-channel DAC and 100G/400G coherent pluggable transceivers use (~800 hours) and control (~200 hours of technical support). The Italian law "codice degli appalti" allows to directly chose the entity for the service "affidamento diretto" below € 140k	€50k
E	CNR	Open-access publication fees	€2.8k
E	UNIVPM	Open-access publication fees, travels to project meetings and conferences, support for the organization of OPTERA's workshop	€21k
E	SSSA	Open-access publication fees	€2.8k

Fig c.11: Details of budget items

2. Information on the new contracts for personnel to be specifically recruited

n°	Associated or principal investigator	Number of expected new contracts	Overall expected time commitment (months)
1.	CIVELLI Stella	2	36
2.	SECONDINI Marco	2	36
3.	BATTAGLIONI Massimo	4	48
4.	GARELLO Roberto	2	36
	Total	10	156

3. Subunit

n°	Associated or principal investigator	Activities carried out by the subunit (if applicable)
1.	CIVELLI Stella	
2.	SECONDINI Marco	
3.	BATTAGLIONI Massimo	
4.	GARELLO Roberto	

4. Financial aspects: costs of each research unit

n°	Research Unit	item A.1	item A.2.1	item B	item C	item D	item E	Total
1.	CIVELLI Stella	233.805	126.000	161.912	0	0	2.800	524.517

2.	SECONDINI Marco	143.375	124.667	120.619	101.000	50.000	2.800	542.461
3.	BATTAGLIONI Massimo	76.500	110.001	83.925	0	15.000	21.073	306.499
4.	GARELLO Roberto	139.125	150.000	130.106	0	0	0	419.231
	Total	592.805	510.668	496.562	101.000	65.000	26.673	1.792.708

N.B. The fields will be filled in automatically

Declarations

☒ I declare that I have completed this application in accordance with the “Guidelines for the evaluation of National Interest Research Projects (PRIN).”

☒ I declare that I have completed this application in accordance with the “Guidelines for the Eligible Expenses”.

☒ Ethics Statement

Upload:

[PRIN-Annex1_EthicsStatementING.pdf](#)

☐ Other documents (if needed)

“The data contained in the application for funding are processed exclusively for carrying out the institutional functions of MUR. The CINECA, Department of Services for MUR, is data controller. The consultation is also permitted to universities, research institutes and institutions (according to respective competence), MUR, CNVR, CdV, and the reviewers in charge of the evaluation.

MUR also has the right to the dissemination of the main economic and scientific data related to the funded projects”.

Date (inserted by system at the closing of the application)